Load-Robust, Physics-Informed Machine Learning for Stator Inter-Turn Fault Diagnosis in Induction Motors: A Leakage-Safe Cascade for Detection, Severity, and Faulted-Phase Identification

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***Abstract*—Stator winding turn-to-turn (inter-turn) short-circuit faults are a leading cause of induction-motor failure, and early, load-robust diagnosis from non-invasive stator-current measurements remains an open problem. Most reported machine-learning (ML) studies achieve near-perfect in-distribution accuracy but never test generalization to unseen operating points, rarely release code, and seldom use leakage-safe evaluation. This paper presents a complete, openly released, physics-informed framework that performs hierarchical diagnosis from the three-phase stator current—fault-onset detection, healthy/faulty classification, ordinal severity estimation (percentage of shorted turns), and faulted-phase identification—and that is evaluated with a strict group-wise protocol and, above all, with *Leave-One-Load-Out* (LOLO) cross-validation. A 357-case PSCAD/EMTDC corpus (f_s = 25 kHz, f_0 = 50 Hz) is segmented into 18,033 labelled windows described by 72 physics-grounded features. LOLO reveals a clean structural result: detection (macro-F1 0.95) and faulted-phase identification (macro-F1 0.997) are intrinsically load-robust because they ride on load-invariant negative-sequence signatures, whereas absolute severity is load-dependent. We then (i) benchmark *eight* models on identical splits—including the tabular foundation model *TabPFN-2.5* and an *xLSTM* sequence network—establishing that, in-distribution, the task is saturated (all feature models within ±0.01 macro-F1) so that the honest discriminator is cross-load behaviour; and (ii) introduce and ablate four load-robustness mechanisms: a sequence-component Stockwell tensor (cross-load severity within-one $0.75 \rightarrow 1.0$), a load-invariance-regularized feature selection, a FiLM load-conditioned multi-task network (cross-load severity $0.93 \rightarrow 0.98$), and a physics-anchored domain adaptation that recovers a simulation-trained detector on a public real-motor dataset ($0.48 \rightarrow 0.98$ macro-F1 with two labelled real samples). The deployed cascade reaches within-one-level accuracy of 0.95 in-distribution and 0.92 across unseen loads, with calibrated detection (expected calibration error 0.021), graceful noise degradation, and 100% fault-onset detection at 0% pre-fault false alarm. A within-hardware evaluation of all eight models on the real motor confirms detection transfers (macro-F1 up to 0.99) and severity is usable; per-window faulted-phase localization is the main gap (macro-F1 0.72 vs. 0.97 in simulation) but per-recording aggregation recovers it to 0.93–0.95—all reported transparently. Finally, leakage-safe Optuna tuning with a cross-load objective and physics-consistent, signature-preserving training-fold augmentation, evaluated under 10-fold grouped cross-validation, further raise cross-load severity (within-one $0.91 \rightarrow 0.94$; in-distribution $0.93 \rightarrow 0.98$; mean absolute error −20%) and detection (LOLO $0.96 \rightarrow 0.98$). A novel block of biomedical anomaly features (signal-complexity, entropy, fractal-dimension, and heart-rate-variability-style cycle-to-cycle measures imported from ECG/EEG analysis) likewise lifts cross-load detection (+0.03) and severity (+0.04) where absolute physics magnitudes do not transfer. All code, splits, and artifacts are released.**

***Index Terms*—Induction motor, inter-turn fault, stator winding, fault diagnosis, severity estimation, machine learning, negative-sequence current, leakage-safe evaluation, foundation model, transfer learning.**

I. Introduction

A. Motivation

INDUCTION motors drive most industrial processes, and stator-winding insulation degradation is among their most common failure modes. An incipient inter-turn (turn-to-turn) short circuit (ITSC) shorts a small fraction of one phase's turns; the circulating fault current accelerates local heating and, if undetected, escalates to a catastrophic phase-to-ground or phase-to-phase fault [1], [2]. Surveys attribute roughly a third of IM failures to stator insulation, and the incipient-to-destructive interval can be short, so an on-line monitor must be both sensitive to small faults and trustworthy enough to trigger maintenance without excessive false alarms. Current-based monitoring (MCSA) is preferred over vibration or thermal sensing because the current is already measured at the drive, needs no extra instrumentation, and is robust to mounting. Three questions must be answered—*is* a fault present, *how severe*, and *which phase*—and the answers must hold up as the mechanical load changes.

B. Literature Review

Penman *et al.* [1] established the ITSC sideband frequencies; Cardoso *et al.* [2] introduced the Extended Park's Vector Approach (EPVA), where the $2f_0$ Park-modulus component scales with winding asymmetry. The negative-sequence current and negative-sequence impedance are widely used asymmetry descriptors [4], [5], with phasor compensation suppressing baseline supply unbalance. Mazzoletti *et al.* [6] use the positive-sequence fifth-harmonic during soft-starter start-up to distinguish ITSC from high-resistance connections; Singh and Shaik [7] combine Stockwell-transform features

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Code, dataset splits, and result artifacts are publicly available at https://github.com/Regal-s/induction-motor-fault-diagnosis.

with SVMs. Data-driven studies feed STFT symmetrical-component amplitudes to SVM/MLP classifiers [9], compare ANN/LSTM/physics-informed networks on a public dataset [10], or ensemble CNNs on spectrograms [11]. Recent architectures (TCN [39], attention [38], selective state-space [21], KAN [22], tabular foundation models) are mostly demonstrated on vibration or generic tabular data rather than three-phase stator current. The recent review [3] flags three systemic weaknesses: suspiciously perfect accuracies from leaky splits, scarce code/data releases, and little validation across operating conditions—the load dependence of the negative-sequence indicator is rarely confronted in a reproducible manner, and operating-condition shift is usually treated as a nuisance to be erased rather than a signal to be exploited.

C. Summary of Contributions

This paper treats ITSC diagnosis as a hierarchical problem (onset → detection → severity → faulted phase) and adopts *Leave-One-Load-Out* (LOLO) evaluation as its central methodological device. Concretely, the delivered system and its contributions are:

- A *leakage-safe evaluation framework*—grouping by operating point and using LOLO as the primary metric—together with a SHAP/Fisher analysis that explains *why* detection and faulted phase are load-invariant while absolute severity is load-dependent. We argue this should be standard practice for the field.
- A *four-stage cascade* on 72 physics-grounded features: onset detection, healthy/faulty detection, ordinal severity estimation, and faulted-phase identification, with calibrated probabilities, conformal severity intervals, and an online onset trigger.
- A *controlled eight-model benchmark* on identical splits—logistic regression, SVM-RBF, k-NN, MLP, random forest, XGBoost, the tabular foundation model *TabPFN-2.5*, and an *xLSTM* sequence network—showing the in-distribution task is saturated and that cross-load behaviour, not in-distribution accuracy, is the meaningful axis of comparison.
- A *novel biomedical anomaly-feature* block—signal-complexity, entropy, fractal-dimension, and heart-rate-variability-style cycle-to-cycle measures imported from ECG/EEG/HRV analysis, computed on the current residual and Park-vector modulus—that complements the physics features and improves cross-load detection (+0.03) and severity (+0.04) where absolute-magnitude features do not transfer.
- Four *load-robustness mechanisms*, each ablated under LOLO: a sequence-component Stockwell tensor (SCST) representation, a load-invariance-regularized feature selection (LIR-mRMR), a phase-coupled FiLM *load-conditioned* multi-task network (PCM-Net) that injects rather than erases the operating point, and physics-anchored domain adaptation (PADA) validated simulation-to-experiment, plus an honest negative result on physics-constrained generative augmentation.
- A *within-hardware evaluation* of all eight models on a real motor (leakage-safe, by recording and by repetition), establishing that detection transfers (macro-F1 up to 0.99) and severity is usable, and transparently identifying faulted-phase localization as the main simulation-to-reality gap.
- A fully *reproducible* release of code, leakage-safe splits, feature tables, and per-model result artifacts.

A unifying theme emerges: the mechanical load is the crux of stator-fault diagnosis, and every proposed mechanism independently improves *cross-load* performance.

II. Proposed Methodology

A. Overview

The system ingests a short three-phase current window and produces, in sequence, a fault-onset estimate, a healthy/faulty decision, and—when faulty—an ordinal severity and the faulted phase. The four questions are nested, have different decision boundaries, and have different load sensitivities; we therefore use a hierarchical, gated architecture where the severity and phase stages are trained and operated only on data they are meant to explain. The thread running through every component—the symmetrical-component representation, the selection criterion, the model conditioning, and the transfer alignment—is load-robustness, which the Leave-One-Load-Out protocol of Section IV verifies. Fig. 1 renders the end-to-end flow.

Pipeline steps. For each 80 ms window $\mathbf{i}(t) = [i_a, i_b, i_c]^\top$: (**S1**) acquire the three currents from the simulation or live ADC, indexed against the operating-point label and the verified region bounds (motor start, load change, fault, clearing; Fig. 2); (**S2**) slide W=2000 samples (4 cycles, 80 ms) at 75% overlap inside the fault interval, labelling the pre-fault region of every faulted record as healthy to obtain matched-load healthy data at every load; (**S3**) divide all three phases by a single per-recording RMS scale (removes load amplitude, preserves inter-phase imbalance); (**S4**) compute symmetrical components $|I_0|, |I_1|, |I_2|$, ratios $|I_2|/|I_1|, |I_0|/|I_1|$, and angles $\angle I_1$, $\angle I_2$ (with sin/cos for wrap-free use) via Clarke and Fortescue; (**S5**) the EPVA severity factor (amplitude of the $2f_0$ line of the Park-vector modulus over its DC), Park-ellipse descriptors, per-phase third-harmonic ratio, THD, RMS / skew / kurtosis and inter-phase differences, plus their $|I_1|$-normalised *load-invariant* n_* variants; (**S6**) 24 biomedical anomaly features (HRV cycle-to-cycle distance, Poincaré SD1/SD2, Hjorth, permutation / spectral / sample / dispersion entropy, Higuchi / Katz / Petrosian fractal dimension) on the residual and Park modulus; (**S7**) group windows by operating point so no recording appears in both train and test, and apply physics-consistent augmentations to the training fold only (§II-F); (**S8**) Stage 0 onset trigger $\rho(k) = |I_2(k)|/|I_1(k)| > \overline{\rho}+6\sigma$ for three consecutive cycles, essentially free, gating the cascade in deployment; (**S9**) Stage 1 class-weighted XGBoost on the 96-feature bank with SHAP audit (ECE 0.021, §IV-I)—if healthy, the cascade terminates; if faulty, S10 and S11 fire in parallel; (**S10**) Stage 2 ordinal regression on $r \in \{0, \ldots, 6\}$ over $\{0.3, 0.5, 1, 2, 3, 4, 5\}$% using the load-invariant subset with the *measured* load (load-aware severity); (**S11**) Stage 3 three-class phase head on the full feature bank, load-invariant by

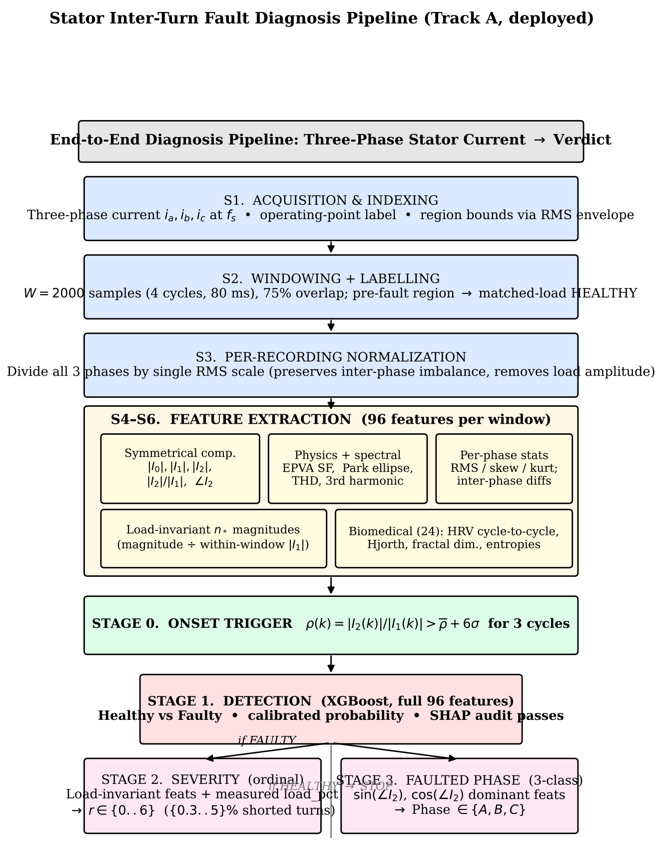

Fig. 1. Block diagram of the deployed diagnosis pipeline (steps S1–S12 of §II-A). Three-phase stator current enters at the top; per-recording normalization (S3) and the 96-feature extractor (S4–S6) feed Stage 0 (onset trigger on the per-cycle $|I_2|/|I_1|$ index) and the gated XGBoost cascade—Stage 1 detection → {Stage 2 severity (load-aware ordinal), Stage 3 faulted phase (3-class)}. Per-recording aggregation (S12) produces the final verdict.

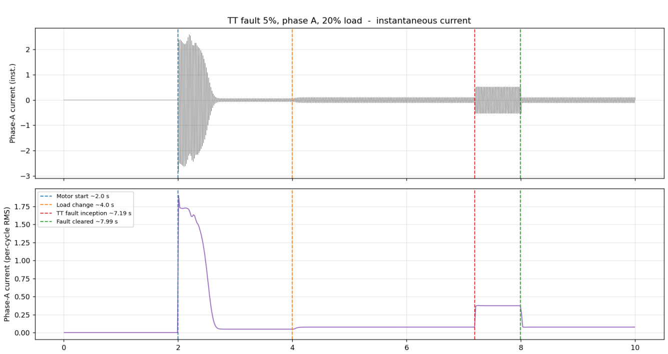

Fig. 2. Phase-A current (top) and per-cycle RMS envelope (bottom) of a faulted record, with motor start (2.0 s), load change (4.0 s), fault inception (7.2 s), and clearing (8.0 s). The fault interval supplies faulty windows; the pre-fault loaded region supplies matched-load healthy windows.

virtue of $\angle I_2$ (§IV-B); (**S12**) per-recording aggregation (majority vote for phase and state, median severity rank) absorbs per-window noise, recovering per-window phase macro-F1 from 0.72 to 0.93–0.95 on real hardware (§IV-J).

B. Feature Families: Four Complementary Views of the Current

A single feature family cannot describe an ITSC fault completely: the fault changes *which* spectral line moves (a frequency-domain phenomenon), *how* the per-phase distributions shift (a statistical phenomenon), *when* the non-stationary signature emerges within a window (a time-frequency phenomenon), and *how regular* the waveform becomes over many cycles (a complexity phenomenon). We therefore compute four complementary feature families on each window and concatenate them into a single 96-dimensional vector (Table II): (i) symmetrical-component and Park-vector descriptors derived from machine theory [2], [1]; (ii) per-phase statistical moments and inter-phase asymmetry features; (iii) a time-frequency sequence-component Stockwell tensor (SCST) [8], [7]; and (iv) signal-complexity / fractal / heart-rate-variability descriptors transferred from biomedical anomaly detection [30], [31], [28], [29], [32], [33], [34]. The first family captures the deterministic asymmetry signature; the second captures empirical distribution shifts; the third localises the signature in time at the operating-point transient; the fourth captures non-spectral irregularity that survives when the absolute spectrum drifts with load.

C. Signal Preprocessing and Symmetrical-Component Descriptors

Preprocessing. Each recording is segmented (§III-C) and each three-phase window is divided by a single per-recording amplitude scale (RMS of the case's healthy region across all three phases), which removes load-induced amplitude variation while preserving the inter-phase imbalance that encodes the fault. Magnitude-based features are additionally divided by the within-window $|I_1|$ to obtain dimensionless load-invariant n_* variants, motivated by the load-stability analysis of §IV-G.

Symmetrical components and Park vector. The Clarke transform [2] underlying the EPVA and Park-vector descriptors is

$$i_\alpha = \tfrac{2}{3}\left(i_a - \tfrac{1}{2}i_b - \tfrac{1}{2}i_c\right), \quad i_\beta = \tfrac{1}{\sqrt{3}}\left(i_b - i_c\right), \tag{1}$$

and the per-window phasors of i_a, i_b, i_c (extracted via a f_0-tuned Goertzel) yield the Fortescue symmetrical components

$$\begin{bmatrix} \mathbf{I}_0 \\ \mathbf{I}_1 \\ \mathbf{I}_2 \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix}\begin{bmatrix} \mathbf{I}_a \\ \mathbf{I}_b \\ \mathbf{I}_c \end{bmatrix}, \quad a = e^{j2\pi/3}. \tag{2}$$

Significance. The negative-sequence ratio $|I_2|/|I_1|$ grows monotonically with the shorted-turn fraction [2], [4], [5] and is the primary severity descriptor; its angle $\angle\mathbf{I}_2$ separates into three $\sim$120°-spaced clusters that identify the faulted phase. The zero-sequence ratio $|I_0|/|I_1|$ captures common-mode imbalance from supply unbalance and grounding asymmetry. A per-case *compensated* value $I_2^c = \mathbf{I}_2 - \overline{\mathbf{I}_2}^{\text{healthy}}$ suppresses the baseline asymmetry due to supply unbalance and isolates the fault contribution. The Extended Park's Vector Approach severity factor [2]

$$\text{SF} = \frac{A(2f_0)}{\text{DC}} \text{ of } |\mathbf{i}_P(t)|, \qquad |\mathbf{i}_P(t)| = \sqrt{i_\alpha^2 + i_\beta^2}, \tag{3}$$

TABLE I
WINDOW-LENGTH TRADE-OFF AT $f_s = 25$ KHZ. Δf IS THE SPECTRAL RESOLUTION OF (4). N_{FAULT} IS THE TYPICAL NUMBER OF NON-OVERLAPPING WINDOWS PER FAULTED RECORDING IN THE 0.8 S FAULT INTERVAL. "EPVA / HARMONICS USABLE" REQUIRES $\Delta f \ll 2f_0$.

Cycles	W	T_{win}	Δf	N_{fault}	EPVA / harm.
1	500	20 ms	50.0 Hz	40	**no** (bins merge)
4	**2000**	**80 ms**	**12.5 Hz**	**10**	**yes (chosen)**
10	5000	200 ms	5.0 Hz	4	yes, but redundant
20	10,000	400 ms	2.5 Hz	2	yes; too few examples

isolates the $2f_0$ component of the Park-vector modulus over its DC and is the strongest single severity scalar in the bank. The 2×2 covariance of (i_α, i_β) gives the Park-ellipse eccentricity $e = \sqrt{1-(b/a)^2}$, axis ratio, and tilt—each zero for a balanced supply and growing monotonically with asymmetry.

Window length: why four cycles. The window length controls the entire feature bank. A W-sample window has spectral resolution

$$\Delta f = \frac{f_s}{W} = \frac{1}{T_{\text{win}}}, \tag{4}$$

so a one-cycle window (W=500, Δf = 50 Hz at 25 kHz) merges DC, f_0, the $2f_0$ EPVA line, and the early Penman sidebands $f_0[1 \pm n(1-s)/p]$ into a few bins—the EPVA severity factor of (3) is then undefined. Four cycles (W=2000, Δf = 12.5 Hz) cleanly separates DC, f_0, $2f_0$, $3f_0$ and the early sidebands, which is the minimum for EPVA, the third-harmonic ratio, THD, and Park-ellipse fitting. It is also the minimum for stable higher-order statistics (skew/kurtosis are high-variance on small samples) and for the HRV-style biomedical descriptors (Poincaré plots require ≥ 4 "heart-beats"). Longer windows (10 or 20 cycles) sacrifice sample efficiency given the 0.8 s fault interval. Table I makes the trade-off explicit; W=2000 is retained throughout.

D. Spectral Descriptors: Harmonic Content

Inter-turn faults introduce specific spectral structure in addition to the fundamental [1]. Two scalars summarise this on each window: the per-phase third-harmonic ratio $R_3 = |\mathbf{I}(3f_0)|/|\mathbf{I}(f_0)|$, sensitive to the 3rd-harmonic asymmetry produced by the inter-turn loop, and the total harmonic distortion THD $= \sqrt{\sum_{k\geq 2}|\mathbf{I}(kf_0)|^2}/|\mathbf{I}(f_0)|$ on per-phase and three-phase mean. The harmonic bins are computed from (f_s, f_0) so the same extractor serves the 25 kHz / 50 Hz simulation and the 1 kHz / 60 Hz experimental data (Section III). We do *not* pre-compute the full Penman sideband family $f_0[1\pm n(1-s)/p]$ [1] because (i) slip s is not directly measured on the public real-motor set and (ii) the SCST representation (§ II-F) recovers the same information in a frequency-localised form that the deep model can exploit.

E. Statistical and Inter-Phase Asymmetry Features

The third family captures *distributional* shifts that the symmetrical-component descriptors do not see. Per-phase root-mean-square, peak, crest factor, standard deviation, skewness, and kurtosis are computed on each of the three phases (6×3 = 18 features); their inter-phase differences (between phases A, B, C, in pairs) *directly encode asymmetry* and are the per-phase analogue of $|I_2|$. A categorical arg max-of-RMS feature flags the dominant-current phase, which empirically resolves the $\angle I_2$ wrap-around for low-severity windows. All magnitude features have $|I_1|$-normalised counterparts (the n_* variants), which Section IV-G shows are the load-stable subset retained by the LIR-mRMR selection criterion. Severity is encoded as an ordinal rank $r \in \{0, \ldots, 6\}$ over $\{0.3, 0.5, 1, 2, 3, 4, 5\}$% so the error metric (within-one-level accuracy, MAE in levels, QWK) respects the ordering of the underlying shorted-turn fraction.

F. Time-Frequency Representation: Sequence-Component Stockwell Tensor

The fourth-family-bridging representation is built from the Stockwell transform [8], a hybrid of the short-time Fourier transform and the continuous wavelet transform that has *frequency-dependent* time resolution: it returns absolute phase, narrower bins at high frequency, and wider bins around the fundamental—precisely the property needed to localise the $2f_0$ EPVA line and the slip-modulated sidebands without the fixed-window compromise of a spectrogram. *SCST construction.* Each phase current is converted to its analytic representation via the Hilbert transform and combined into instantaneous symmetrical-component streams $i_0(t), i_1(t), i_2(t)$ using the Fortescue matrix of (3); a band-limited Stockwell transform is applied to each stream over the diagnostic band $[0.5f_0, 10f_0]$, and the three magnitude maps $|S_0|, |S_1|, |S_2|$ are stacked into a single 3-channel time-frequency tensor (the SCST). *Significance.* The negative-sequence channel $|S_2(t,f)|$ carries the load-invariant inter-turn signature localised in both time and frequency, which is what a 2-D CNN can exploit (Section IV-E, where the SCST raises cross-load severity within-one from 0.75 to 1.0). Because the symmetrical-component *magnitudes* are phase-agnostic by construction, the magnitude tensor cannot localise the faulted phase; phase identification therefore continues to use the negative-sequence *angle* as in § II-C.

G. Biomedical Anomaly Features

The fourth family captures *non-spectral irregularity*: how predictable the waveform is, regardless of which spectral lines exist. We transfer features routinely used in ECG/EEG anomaly detection to the motor-current setting—a view absent from the ITSC literature, which treats the current as a deterministic sum of sinusoids and asks *which line moved* rather than *how regular* the waveform is. The premise is that an inter-turn short makes the waveform less regular within and across cycles, which signal-complexity measures quantify directly. Crucially, these are computed on the *residual* (current minus its fitted fundamental) and on the Park-vector modulus, not on the raw current whose clean fundamental is trivially predictable. Three sub-groups are used.

(a) HRV-on-current. The most physically faithful import, since an ITSC *is* cycle-to-cycle irregularity: cycle-to-cycle waveform distance to the median cycle, and Poincaré SD1, SD2, SD1/SD2 [33], [34] of the per-cycle RMS series. Each

TABLE II
THE 96-FEATURE PER-WINDOW BANK, ORGANISED BY THE FOUR COMPLEMENTARY FAMILIES OF § II-B. THE FIRST THREE FAMILIES (72 FEATURES) CONSTITUTE THE PHYSICS+STATISTICAL BANK USED BY THE DEPLOYED XGBOOST CASCADE; THE FOURTH FAMILY (24 BIOMEDICAL FEATURES) IS APPENDED IN THE LOAD-ROBUSTNESS ABLATION (TABLE XV). ALL QUANTITIES ARE COMPUTED PER WINDOW; MAGNITUDE FAMILIES HAVE $|I_1|$-NORMALISED COUNTERPARTS (n_* VARIANTS).

Family / group	#	Features	What it captures about ITSC	Refs
(i) Symmetrical-component / Park	24	$\lvert I_0\rvert, \lvert I_1\rvert, \lvert I_2\rvert$; ratios $\lvert I_2\rvert/\lvert I_1\rvert, \lvert I_0\rvert/\lvert I_1\rvert$; angles $\angle I_1, \angle I_2$ with sin/cos; compensated I_2^c (abs, ratio, angle); EPVA severity factor; Park-modulus mean/std; Park-ellipse major, minor, eccentricity, axis ratio, tilt	*Deterministic asymmetry signature.* $\lvert I_2\rvert/\lvert I_1\rvert$ scales monotonically with μ (severity); $\angle I_2$ in three 120° clusters identifies the faulted phase; $\lvert I_0\rvert/\lvert I_1\rvert$ flags supply unbalance vs. ITSC; EPVA isolates the $2f_0$ Park-modulus line, the strongest single severity scalar	[2], [4], [5], [6]
(ii) Spectral / harmonic	6	3rd-harmonic ratio R_3 (per-phase $\times 3$, mean); THD (per-phase, mean)	*Harmonic distortion of the fault loop.* The inter-turn loop produces 3rd-order asymmetry; THD captures aggregate spectral spread	[1], [7]
(iii) Statistical / inter-phase	42	RMS, peak, crest, std, skew, kurtosis (per phase, 6×3); pairwise inter-phase differences (RMS/peak/crest/std $\times 3$ pairs); arg max-of-RMS categorical; load-invariant n_* variants of all magnitude features	*Per-phase distributional shifts.* Higher-order moments are sensitive to the non-sinusoidal distortion in the faulted phase; inter-phase differences encode asymmetry directly without sequence-domain conversion	[9], [10]
Subtotal (deployed XGBoost cascade)				**72**
(iv) Biomedical anomaly	24	HRV cycle-to-cycle waveform distance and Poincaré SD1/SD2 of the per-cycle RMS; Hjorth mobility and complexity; permutation, spectral, sample, dispersion entropy; Higuchi, Katz, Petrosian fractal dimension—all computed on the *residual* signal (current − fundamental) and on the Park-vector modulus	*Non-spectral irregularity / complexity.* Captures how regular the waveform is rather than which lines moved; the residual is anomaly-rich because the fundamental is removed; amplitude-scale-free measures transfer across loads (§IV-G)	[33], [34], [30], [29], [28], [3

fundamental cycle is treated as a heartbeat; the per-cycle RMS series is the direct analogue of the inter-beat-interval series in HRV.

(b) Complexity scalars. Hjorth mobility and complexity [30] (amplitude-scale-free time-domain derivatives that quantify the rate of zero-crossings and the smoothness of the derivative), permutation entropy [29] (rank-based, robust to monotone amplitude transforms), spectral entropy (Shannon entropy of the normalised power spectrum, captures spectral flatness), sample entropy [28] (self-similarity), and dispersion entropy [32] (quantises the signal into classes and measures the entropy of the resulting symbol sequence).

(c) Fractal dimension. Higuchi [31], Katz, and Petrosian fractal dimensions, which quantify the self-affine roughness of the residual on multiple scales.

Why these specifically. The complexity and fractal members are amplitude-scale-free by construction, which the LOLO ablation of Table XV confirms transfers across loads better than absolute spectral features. The block adds no heavy computation (no $O(N^2)$ recurrence matrices) and is appended to the 72-feature physics+statistical bank for a total of 96 per-window features.

H. Per-Task Discriminative Features

The three diagnostic tasks (detection, severity, faulted phase) do *not* rely on the same features—a fact directly confirmed by the SHAP attribution of § IV-G, the load-stability analysis of the LIR-mRMR selection of § IV-G, and the biomedical ablation of § IV-F. Table III maps the dominant discriminators per task, distinguishing the in-distribution drivers from the cross-load (LOLO) ones, so the reader can see at a glance which family carries which task.

Detection (§ IV-B). In-distribution detection is driven by physics: the Park-vector major axis and eccentricity, the negative-sequence ratio $|I_2|/|I_1|$, the zero-sequence ratio $|I_0|/|I_1|$, the inter-phase RMS differences, and the EPVA severity factor lead the SHAP attribution—an inter-turn short bends the Park trajectory into a perceptible ellipse and breaks the symmetric-component balance, both reflected in these scalars. *Cross-load* detection is the one cell where the biomedical features measurably *add* discriminative power: Table XV reports a +0.030 LOLO macro-F1 gain when appended, traced (§ IV-G, Fig. 10b) to the Park-modulus Hjorth parameters and the residual permutation/dispersion entropy—all amplitude-scale-free and therefore robust to the load-induced amplitude shift that displaces the absolute physics magnitudes.

Severity assessment (§ IV-B, § IV-C). The strongest in-distribution severity discriminator is a *single* physics scalar—the EPVA severity factor SF of (4), with $|I_2|/|I_1|$ and the Park-ellipse eccentricity close behind. Because all three scale with the operating point, these are also the features most affected by the LOLO load shift; the load-aware fix of § IV-C (supplying the measured load) and the biomedical features together close the gap. Among the biomedical block, the residual Higuchi/Petrosian fractal dimensions and the dispersion entropy carry the most severity weight (Fig. 10b), contributing the +0.037 LOLO within-one gain of Table XV.

Faulted-phase identification (§ IV-B). This is the task that uses essentially *one* feature: the negative-sequence angle $\angle I_2$ (encoded as sin/cos for wrap-free use), which is by construction load-invariant and clusters in three 120°-spaced sectors. The per-phase arg max-of-RMS and the inter-phase RMS differences play a backup role for low-severity windows where $\angle I_2$ is noisy. The biomedical block contributes essentially

TABLE III

Per-task dominant discriminative features. "In-dist." columns are SHAP top-ranked features under StratifiedGroupKFold; "LOLO" columns are the features whose presence most improves cross-load accuracy. *Italicised* entries are amplitude-scale-free. Biomedical features are decisive for cross-load detection and severity but contribute essentially nothing to phase identification.

Task	In-distribution drivers (physics-dominated)	LOLO drivers (load-stable)
Detection	Park major axis; Park eccentricity; $\lvert I_2\rvert/\lvert I_1\rvert$; $\lvert I_0\rvert/\lvert I_1\rvert$; inter-phase RMS diffs; EPVA SF	*Park-modulus Hjorth* (mobility, complexity); *residual permutation entropy*; *dispersion entropy*; LIR-mRMR load-invariant subset (+0.030 LOLO F1)
Severity	EPVA SF (dominant); $\lvert I_2\rvert/\lvert I_1\rvert$; Park-ellipse eccentricity; $\lvert I_2^c\rvert/\lvert I_1\rvert$	Measured load + load-invariant n_* features (§ IV-C, 0.73 → 0.95); *Higuchi / Petrosian FD*; *dispersion entropy*; *Hjorth complexity* (+0.037 LOLO within-1)
Phase ID	$\angle I_2$ (sin, cos, *essentially the single feature*); arg max-of-RMS; per-phase RMS differences	Same as in-distribution—*$\angle I_2$ is intrinsically load-invariant*; biomedical contribution +0.000 (Table XV). Per-recording majority vote of $\angle I_2$ closes the residual per-window noise on real hardware (§ IV-J)

nothing here (+0.000 LOLO macro-F1, Table XV)—phase identification is a pure inter-phase asymmetry problem that the symmetrical-component angle already solves, so adding signal-complexity scalars to a phase-agnostic problem has no effect, exactly as predicted.

This task-specific feature structure is the reason the deployed pipeline is a *cascade* rather than a single multi-class model: each stage selects only the features that discriminate *its* task. Stage 1 uses all 96 features (so the biomedical block contributes to cross-load detection); Stage 2 uses the load-invariant subset + measured load (so the deterministic-asymmetry features that drift with load are excluded); Stage 3 uses the full bank but is effectively driven by $\angle I_2$ (so load-invariance is automatic). Section IV-G visualises the per-task feature rankings; the present subsection summarises them as a concise design constraint on the cascade.

I. Data Augmentation and Feature Selection

Feature selection. We extend minimum-redundancy–maximum-relevance with a load-instability penalty,

$$J(f) = \mathrm{Rel}(f) - \lambda\,\mathrm{Red}(f) - \gamma\,\mathrm{LoadInstability}(f), \quad (5)$$

where $\mathrm{LoadInstability}(f)$ is the across-load drift of the feature's class-conditional mean normalized by its global spread (γ=0 recovers plain mRMR). LIR-mRMR selects dimensionless, load-stable features and, on the load-dependent severity task, generalizes better at compact size (Section IV-E). *Augmentation.* For the deployed pipeline we apply, to the *training fold only*, a set of physics-consistent augmentations selected from a survey of time-series and signature-guided methods [26], [27]: joint amplitude (load) scaling, per-channel SNR jitter, mild magnitude-warp, phase-coherent time shift, and—for the faulted-phase task—exact faulted-phase rotation (a cyclic channel permutation with the negative-sequence angle rotated ∓120°). Every amplitude operation acts *jointly* on the three phases so the inter-phase imbalance (the fault signature) and Kirchhoff balance $i_a + i_b + i_c = 0$ are preserved; a current-balance reject filter guards this, and signature-corrupting operations (per-channel scaling, segment permutation, phase-spectrum randomization) are excluded. Augmented windows are re-featurised through the same extractor. We separately study a conditional generator of three-phase current with a hard Kirchhoff projection and a soft negative-sequence-versus-severity loss; the hard constraint is enforced exactly, with an honest negative result on controllability reported in Section IV-E. Hyperparameters of the gradient-boosted models are selected by Optuna TPE [25] with a *cross-load* inner objective (leave-one-load-out over the training loads, the model-selection principle of domain generalization [24]), with early stopping on an inner held-out load.

J. Models: Gradient Boosting, Foundation Model, and Deep Networks

Gradient-boosted trees (XGBoost [12]) implement the three stages: detection (class-weighted binary), severity (ordinal regression on the rank), and faulted phase (three-class). Stages 1 and 3 use the full feature set; the severity stage uses the load-invariant subset and the *measured* load. The detector gates the downstream stages: windows predicted faulty are passed to the severity and phase models. To place the cascade in context we evaluate, on identical splits, six classical feature classifiers and two recent learners on the same 72 features or raw windows: *TabPFN-2.5* [13], a pretrained tabular foundation model that performs in-context Bayesian inference over a feature table, and *xLSTM* [14], an extended LSTM with exponential gating and a stabilizer state applied to the raw three-phase current. As a load-conditioned deep alternative we develop *PCM-Net*, a phase-coupled, multi-task network with a dilated-temporal backbone (a compact stand-in for a selective state-space model) and *feature-wise linear modulation* (FiLM) [35] driven by the measured operating point, so one model spans loads instead of erasing the condition; a current-balance physics term regularizes training. A fault-onset stage thresholds a per-cycle negative-sequence index $\rho(k) = |I_2(k)|/|I_1(k)|$ against a trailing pre-fault reference, $\rho(k) > \overline{\rho} + 6\sigma$ for three consecutive cycles. Detection probabilities are assessed with expected calibration error and the Brier score [18], and severity uncertainty with conformalized quantile regression [16], [17]. Fig. 3 contrasts the two tracks side by side: Track A (deployed) takes the 96 physics/biomedical features into the gated XGBoost cascade with a per-recording aggregator, while Track B (alternative) feeds the raw three-phase window through the PCM-Net stem, phase-coupling attention, FiLM-conditioned backbone, and per-task heads.

K. Transfer Learning

To deploy a simulation-trained model on a real machine we use physics-anchored domain adaptation (PADA). A naively

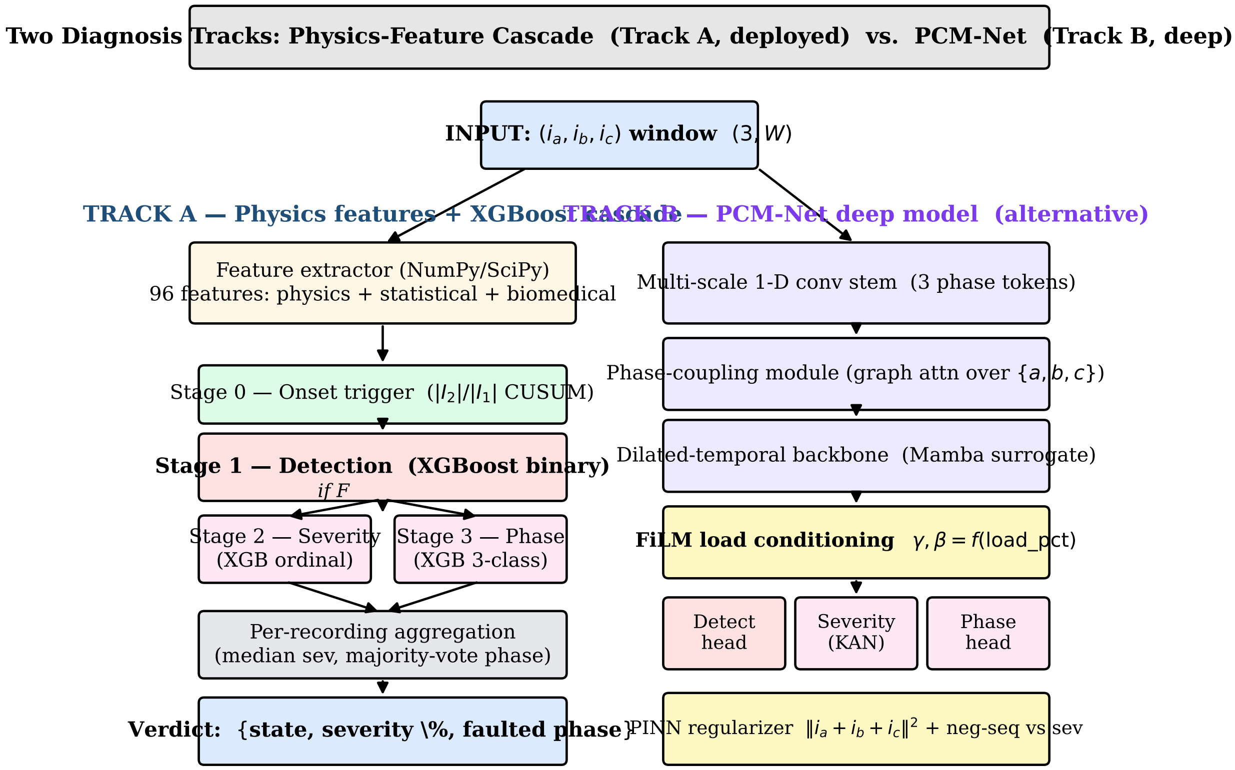

Fig. 3. The two diagnosis tracks evaluated in this work. **Track A** (left, the *deployed* cascade): NumPy/SciPy feature extractor → Stage 0 onset → Stage 1 XGBoost detector → {Stage 2 ordinal-severity head, Stage 3 3-class phase head} → per-recording aggregation. **Track B** (right, the deep alternative): raw $(3, W)$ window → multi-scale 1-D conv stem → phase-coupling graph attention → dilated-temporal (Mamba-surrogate) backbone → FiLM load conditioning → {detect, severity (KAN), phase} heads with a current-balance PINN regularizer. Both tracks accept the same input and emit the same verdict; Section IV-D reports their compared performance.

transferred model collapses under domain shift; we therefore align feature distributions across domains, anchoring the alignment on the domain-stable physics features (negative-sequence ratio, angles, $|I_1|$-normalized magnitudes), and optionally calibrate with a few labelled target repetitions. Section IV-E quantifies the recovery on a public real-motor dataset.

III. Experimental Setup and Dataset Preparation

Two complementary datasets are used in this work. A large *simulation* corpus from PSCAD/EMTDC supports the in-distribution, leave-one-load-out, and ablation studies in §IV-B–H; a public *real-motor* dataset [36] supports within-hardware validation (§IV-J), simulation-to-experiment transfer learning (§IV-E), and the real-time edge deployment of §IV-K. The two sources are deliberately heterogeneous in machine, line frequency, sampling rate, severity scale, sensing chain, and the presence of a healthy baseline; this section documents each, the construction of the test-case matrix, and the window segmentation and labelling protocol that feeds the models.

A. Simulation Dataset: PSCAD/EMTDC Stator Inter-Turn Faults

Machine and fault model. A three-phase squirrel-cage induction motor is modelled in PSCAD/EMTDC at line frequency $f_0 = 50$ Hz. The stator inter-turn fault is implemented as a *tapped winding*: a controlled fraction μ of one phase's turns is tapped out and short-circuited through a fault impedance, so that the model exposes both magnitude and phase asymmetry while remaining electromagnetically consistent with the unfaulted stator. The three line currents (i_a, i_b, i_c) are exported as raw time-series; no filtering or spectral shaping is applied before the feature pipeline so that the diagnosis operates on the raw measurement.

Sampling and run length. Each run is sampled at f_s = 25 kHz, giving 500 samples per fundamental cycle and a 40 μs time step. Faulted runs are 10 s long (250,000 samples); healthy runs are 5 s long (125,000 samples). Three multi-channel `.out` files per case carry the full measurement bus; the diagnostically relevant channels are the three line currents in kA, used directly throughout this paper.

Duty-cycle schedule. Each faulted run follows a fixed event schedule that mimics a realistic motor duty cycle (Fig. 2): the motor is energized at 2.0 s (start-up inrush settles by ∼ 2.7 s), the mechanical load is applied at 4.0 s (skipped for the no-load variant), the inter-turn short is inserted at 7.2 **s**, and the fault is cleared ≈ 0.8 s later. This single-run schedule furnishes both a faulted segment (samples 180,000–200,000) and a *matched-load* healthy segment (the pre-fault region, samples ∼110,000–178,000), the latter providing load-matched healthy data and a per-recording baseline for negative-sequence compensation (§II-B).

TABLE IV
PSCAD/EMTDC SIMULATION CORPUS—ENUMERATION OF THE OPERATING-POINT TEST MATRIX AND RECORDING COUNTS.

Axis	Levels
Fault type	Stator inter-turn (tapped winding)
Severity μ (% shorted turns)	$\{0.3, 0.5, 1, 2, 3, 4, 5\}$
Faulted phase	$\{A, B, C\}$
Load (% rated)	$\{\mathrm{NL}, 20, 40, 60, 80, 100\}$
Fault inception t_f (s)	$\{7.200, 7.202, \ldots, 7.220\}$ (11 variants)
Faulted recordings (7×3×6×11 − unused cells)	336
Healthy recordings	21
Total simulation recordings	**357**
Sampling f_s	25 kHz (500 samples/cycle)
Run length (faulted / healthy)	10 s / 5 s
Operating-point groups (CV grouping)	147

TABLE V
REAL-MOTOR DATASET [36]—ENUMERATION OF THE TEST MATRIX AND RECORDING COUNTS.

Axis	Levels
Motor	0.75 hp squirrel-cage IM, f_0 = 60 Hz
Sampling f_s	1 kHz
Sensing chain	Split-core current transformers
Mechanical load	None (no-load operation)
Severity (% shorted turns)	$\{10, 20, 30, 40\}$
Faulted phase	$\{A, B, C\}$
Healthy condition	{HLT}
Repetitions per condition	{01, 02, 03, 04, **05**}
Total distinct conditions	13 (1 healthy + 12 faulted)
Total recordings	65
Labelled windows	1,235
Held-out test repetition	rep 05 (13 recordings)

Operating-point matrix. The corpus enumerates a four-axis matrix (Table IV): severity $\mu \in \{0.3, 0.5, 1, 2, 3, 4, 5\}$% shorted turns; faulted phase $\in \{A, B, C\}$; load $\in \{\mathrm{NL}, 20, 40, 60, 80, 100\}$%; and fault inception $t_f \in \{7.200, 7.202, \ldots, 7.220\}$ s (11 variants per $(\mu, \mathrm{phase}, \mathrm{load})$ triple, sweeping the inception over one half-cycle of f_0 to randomise the fault-initiation angle). A healthy subset spans the same load range. Total: 336 faulted + 21 healthy = 357 recordings → 18,033 labelled windows (Table VI).

B. Real-Motor Experimental Dataset

The real-motor validation uses a public laboratory dataset [36] acquired on a 0.75 hp squirrel-cage IM at f_0 = 60 Hz, f_s = 1 kHz, no-load, with three-phase current sensed by split-core CTs. The same machine produces both classes via external stator tappings (clean sensing-chain symmetry). The matrix (Table V) spans severity $\{10, 20, 30, 40\}$%, faulted phase $\{A, B, C\}$, and a healthy class, over 5 repetitions per condition: 13 conditions ×5 reps = 65 recordings → 1,235 labelled windows. Repetition 05 is held out throughout the paper; train splits use 01–04.

The two datasets are complementary: PSCAD carries the load axis (essential for LOLO and the load-aware-severity story of §IV-C); the real-motor set carries the sensing and machine realism but no load axis. Together they support in-distribution evaluation, cross-load LOLO, within-hardware validation, simulation-to-experiment adaptation, and the real-time edge deployment of §IV-K.

TABLE VI
SAMPLE DISTRIBUTION ACROSS BOTH DATASETS AFTER WINDOWING. WINDOWS ARE GROUPED BY OPERATING POINT (THE LEAKAGE-SAFE GROUPING UNIT, 147 GROUPS FOR THE SIMULATION CORPUS AND 13 FOR THE REAL-MOTOR CORPUS); INCEPTION-TIME VARIANTS AND PER-CONDITION REPETITIONS ARE BOUND INTO THE SAME GROUP AS THE BASE CASE.

	Simulation (PSCAD/EMTDC)	Real motor [36]
Recordings	357	65
– healthy	21	5 (HLT × 5 reps)
– faulted	336	60 (12 cond × 5 reps
Labelled windows	**18,033**	**1,235**
– healthy	6,273	95
– faulted	11,760	1,140
Window length W	2,000 samples (80 ms)	500 samples (500 m
Stride (in / out of fault)	500 / 4,000	50 / –
Severity balance	1,680 per μ level	~ 285 per μ level
Phase balance	3,920 per phase	380 per phase
CV grouping unit	operating-point (severity × phase × load)	condition × rep
Number of CV groups	147	13
Held-out test partition	per protocol (GroupKFold or LOLO)	repetition 05

C. Window Segmentation, Labelling, and Sample Distribution

Window length is W = 2000 samples for the simulation (4 cycles at 25 kHz, 80 ms) and W = 500 for the real-motor set (500 ms at 1 kHz)—the smallest window for which the EPVA $2f_0$ line, the third-harmonic ratio, the Park ellipse, and higher-order per-phase statistics are all well posed ($\Delta f = f_s/W = 12.5$ Hz at 25 kHz cleanly separates DC, f_0, $2f_0$ and early Penman sidebands). Inside the 0.8 s simulation fault interval the stride is 500 samples (75% overlap) to extract enough labelled faulted windows; outside it (and on the real-motor set) a sparser stride balances the corpus without inflating it. The pre-fault region of every *faulted* run is labelled healthy at its load level, which (i) furnishes matched-load healthy data at every load (preventing the detector from learning load as a class proxy) and (ii) supplies the on-recording healthy baseline used by per-recording normalisation (§II-B) and the compensated I_2^c. Each window is divided by a single per-recording amplitude scale (RMS of the healthy region across the three phases), removing load amplitude while preserving inter-phase imbalance; magnitude features are additionally divided by the within-window $|I_1|$ to give the load-invariant n_* variants. The simulation corpus is balanced across severity (1,680 per level) and faulted phase (3,920 per phase); the healthy class has 6,273 windows (6,138 matched-load pre-fault + 135 from dedicated healthy runs). Table VI gives the full breakdown.

All inception variants of an operating point and all repetitions of a real-motor condition are bound into a single CV group so no window from a single recording can appear in both train and test (147 sim groups, 13 real). The two protocols are Stratified Group k-Fold (k=5) for in-distribution estimation, and *Leave-One -Load-Out* (LOLO)—all windows at one load form the test set in turn—as the primary metric (§IV-A).

IV. Performance Evaluation

This section evaluates the proposed framework end-to-end. §IV-A specifies the implementation, splits, and metrics; §IV-B reports the main cascade performance in-distribution and under the primary cross-load (LOLO) protocol; §IV-C analyses *why* the three tasks differ under LOLO and introduces the load-aware severity fix; §IV-D compares eight model classes on identical splits and tests cross-model significance; §IV-E–H isolate the contribution of every novel component (load-robustness mechanisms, biomedical features, load-invariant feature selection, and physics-consistent augmentation/tuning) through controlled ablations; §IV-I quantifies probabilistic calibration, noise robustness, and online fault-onset detection; §IV-J validates the cascade within-hardware on a publicly released real-motor dataset; §IV-K reports the measured end-to-end real-time validation on a Jetson Orin Nano edge module; and §IV-L positions the framework relative to the literature and lists the residual limitations.

A. Implementation and Evaluation Protocol

Features are computed in Python (NumPy/Pandas); XGBoost uses 400–500 trees, depth 5, η=0.05, subsample/colsample 0.8; deep models use PyTorch and TabPFN-2.5 runs on CPU. Reported metrics are macro-F1, severity within-one-level accuracy, MAE in levels, quadratic-weighted kappa (QWK), and per-phase F1, on out-of-fold predictions averaged over folds. Stages 2–3 are trained on faulty windows but evaluated on the windows the detector predicts faulty, so cascade numbers reflect realistic error propagation; predictions are also aggregated per recording (median severity, majority-vote phase). Scalers are fit on the training partition only, every benchmark model sees identical folds, no per-model tuning is done on test folds, and all randomised components use fixed seeds.

B. Main Cascade Performance: In-Distribution and Cross-Load

Table VII summarises the deployed cascade. *The central finding is structural rather than numerical:* in-distribution every stage exceeds 0.97, but only LOLO separates the three tasks into those that genuinely generalise across operating points (detection 0.948, faulted phase 0.997) and the one that does not without intervention (absolute severity). The detection ROC has AUC 0.994 (Fig. 4) and the faulted-phase confusion (Fig. 5) is near-diagonal with phase C perfect. End-to-end, the gated cascade reaches within-one accuracy 0.949 in-distribution and 0.916 cross-load (per-recording 0.927 / 0.918; Fig. 6). Faulted phase is the only metric for which cross-load *exceeds* in-distribution, a direct consequence of its load-invariance.

C. Cross-Load Analysis and the Load-Aware Severity Fix

SHAP attribution shows detection and phase ride on the *negative-sequence* ratio and angle and on inter-phase RMS differences—all load-invariant ratios—whereas an initial severity model built on absolute magnitudes *collapsed* at the no-load extreme under LOLO (within-one 0.39), because the $|I_2|$ magnitude that encodes severity scales with the operating point. Two complementary remedies recover it (Table VIII). First, restricting severity to dimensionless, load-invariant features and $|I_1|$-normalized magnitudes lifts the no-load case from 0.39 to 0.80. Second—and decisively—supplying the *measured* load (a sensor input, not a label) raises pooled LOLO within-one from 0.73 to 0.95, since most held-out loads are interpolated between neighbours. Severity errors are then almost entirely between adjacent levels (Fig. 7, QWK 0.987), which validates the ordinal treatment. The per-held-out-load breakdown (Table IX) localizes the residual difficulty to the light-load extremes (no-load and 20%), where a low-severity fault current most overlaps the healthy distribution—a difficulty visible *only* under LOLO.

TABLE VII
Deployed cascade: per-stage and end-to-end performance (mean over folds).

Stage	Metric	Group k-Fold	LOLO (primary)
1 Detection	macro-F1 / ROC-AUC	0.977 / 0.994	0.948
2 Severity (load-aware)	within-1 / QWK	0.997 / 0.987	0.946
3 Phase ID	macro-F1	0.972	**0.997**
End-to-end	within-1 / per-record	0.949 / 0.927	0.916 / 0.918

TABLE VIII
Severity within-one-level accuracy under feature choice. The load-aware model is the deployed configuration.

Severity feature set	Group k-Fold	LOLO	No-load
All features (magnitudes incl.)	0.993	0.887	0.39
Load-invariant + $|I_1|$-norm	0.990	0.729	0.80
+ measured load (proposed)	**0.997**	**0.946**	**0.95**

TABLE IX
Per-held-out-load detection (LOLO, macro-F1). Difficulty concentrates at light loads.

Hold-out load	NL	20%	40%	60%	80%	100%
Detection F1	0.885	0.749	1.000	1.000	1.000	1.000

D. Comparative Model Evaluation

Table X and Fig. 8 benchmark eight models on *identical* leakage-safe splits, spanning classical feature classifiers, a tabular foundation model, and a sequence deep network. The first observation is that the *in-distribution task is saturated*: the six feature classifiers fall within ±0.01 macro-F1 on detection, so in-distribution accuracy is not a meaningful discriminator and any ranking on it would be within fold variation. The honest comparison is therefore cross-load, where three results stand out. (1) *TabPFN-2.5* is the strongest detector and uniquely load-robust, scoring macro-F1 0.979 both in-distribution and cross-load, vs. XGBoost 0.976 → 0.948; it also has the sharpest in-distribution severity (0.997 within-one, MAE 0.033) and degrades gracefully across loads (0.969). (2) Severity separates models: tree ensembles extrapolate poorly (RF LOLO within-one 0.688) while the MLP and SVM remain robust (0.982,

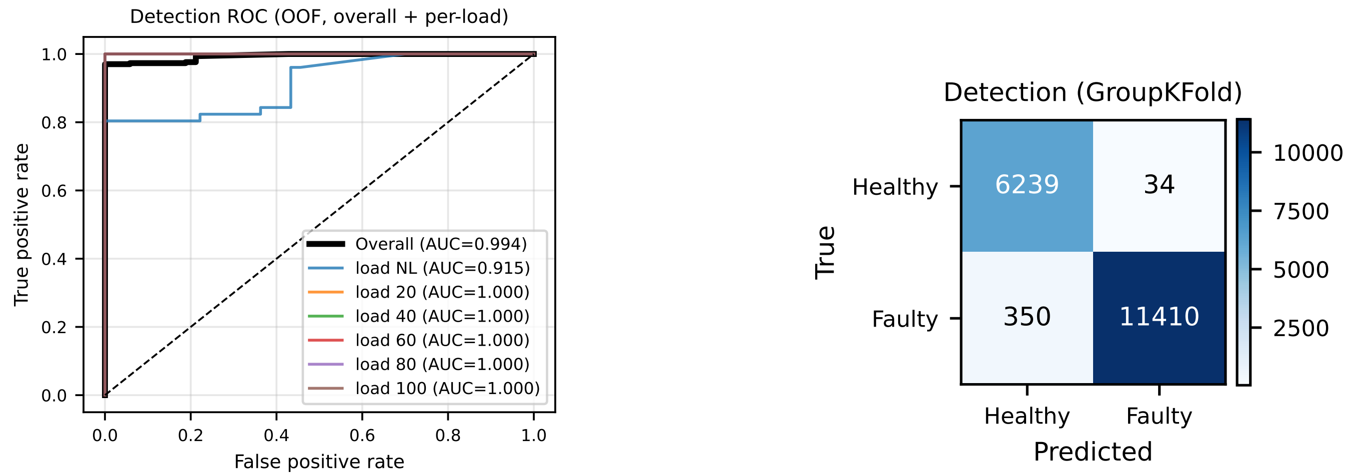

Fig. 4. Detection. Left: receiver operating characteristic, overall and per held-out load (overall AUC = 0.994). Right: confusion matrix (group k-fold, out-of-fold).

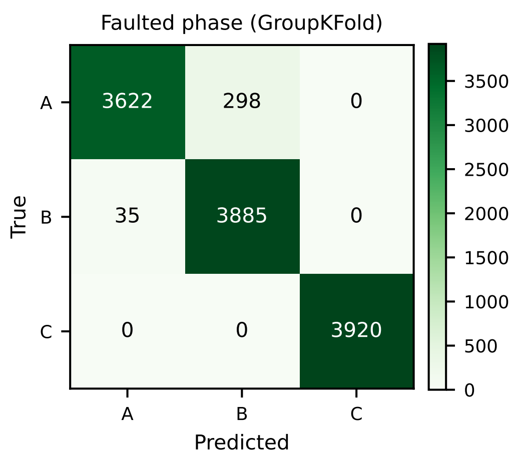

Fig. 5. Faulted-phase confusion (group k-fold, out-of-fold). Phase C is perfect; only minor A↔B confusion remains. Cross-load (LOLO) phase accuracy is 0.997.

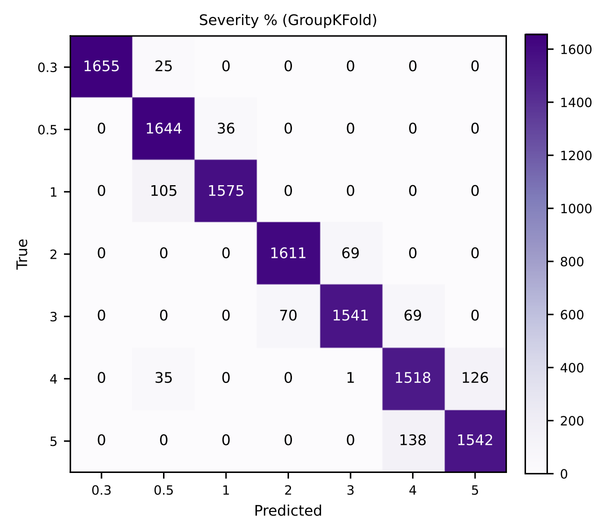

Fig. 7. Severity confusion (group k-fold). Errors are almost entirely between adjacent levels, justifying the ordinal treatment and the within-one-level metric (QWK = 0.987).

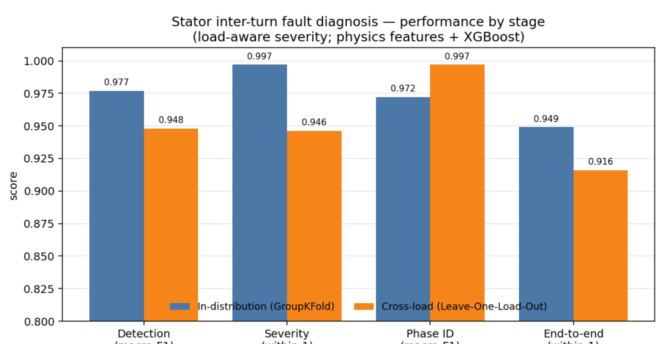

Fig. 6. Per-stage and end-to-end performance, in-distribution (group k-fold) vs. cross-load (LOLO). Phase identification is the only metric for which cross-load exceeds in-distribution, reflecting its load-invariance.

0.975). (3) Faulted phase is load-robust for nearly every model (LOLO macro-F1 ≥ 0.96), confirming the negative-sequence-angle signature is model-agnostic. Raw-signal *xLSTM* trails the feature learners on every task (LOLO 0.890/0.908/0.961) yet improves on the earlier raw-signal deep baselines (Table XII), reaffirming the data-efficiency of physical features. The pipeline is lightweight ($\approx$ 0.34 ms per window on a desktop CPU, Fig. 9) and TabPFN-2.5 needs no task-specific training—the practitioner picks foundation-model robustness or gradient-boosted speed without sacrificing accuracy.

Best model per task. Table XI distils the per-cell winners: *TabPFN-2.5 is the only model top-tier on all three tasks*—best (tied) cross-load detector, lowest in-distribution severity MAE (0.033 vs. next-best 0.095), and within 0.006 macro-F1 of the best phase score. MLP edges TabPFN on LOLO severity (0.982 vs. 0.969); XGBoost and RF tie on LOLO phase. The deployed XGBoost cascade can therefore be upgraded in place by swapping Stage 1 and the in-distribution Stage 2 head for

TABLE X

FULL MODEL BENCHMARK ON IDENTICAL LEAKAGE-SAFE SPLITS: STRATIFIEDGROUPKFOLD (GK, IN-DISTRIBUTION) AND LEAVE-ONE-LOAD-OUT (LOLO, CROSS-LOAD, PRIMARY). DETECTION AND PHASE REPORT MACRO-F1; SEVERITY REPORTS WITHIN-ONE-LEVEL ACCURACY. BEST PER COLUMN IN BOLD. TABPFN-2.5 AND XLSTM ARE ADDED IN THIS WORK.

	Detection (macro-F1)		Severity (within-1)		Phase (macro-F1)	
Model	GK	LOLO	GK	LOLO	GK	LOLO
Logistic Regression	0.978	0.978	**0.997**	0.971	0.970	0.920
SVM-RBF	0.978	0.977	0.991	0.975	**0.984**	0.967
k-NN	0.977	0.977	0.937	0.942	0.982	0.981
MLP	0.978	0.978	**0.997**	**0.982**	0.956	0.981
Random Forest	0.978	**0.979**	0.985	0.688	0.965	**0.997**
XGBoost	0.976	0.948	0.990	0.943	0.976	**0.997**
TabPFN-2.5	**0.979**	**0.979**	**0.997**	0.969	0.968	0.991
xLSTM	0.957	0.890	0.982	0.908	0.921	0.961

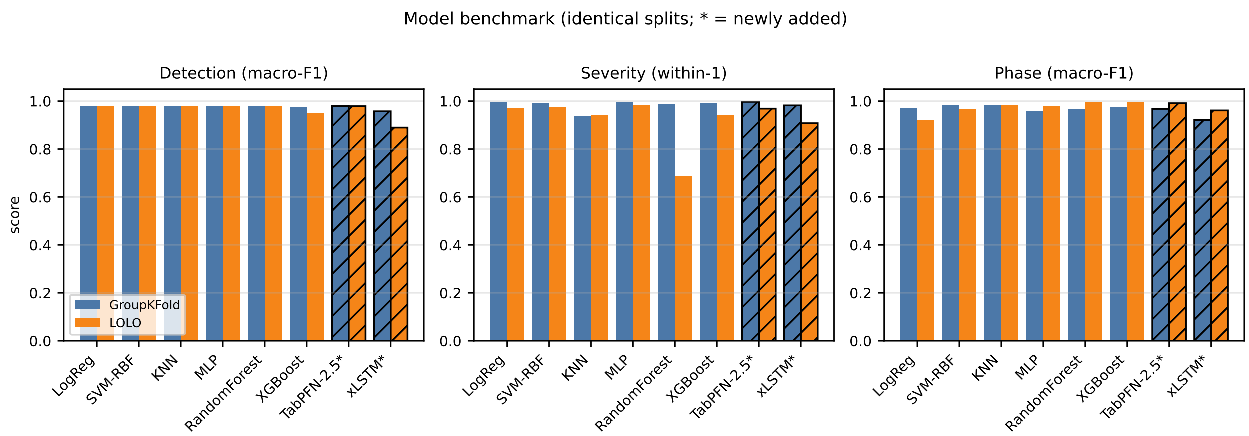

Fig. 8. Model benchmark across the three diagnostic tasks under both protocols (GroupKFold vs. LOLO). Hatched bars denote the two models added in this work (TabPFN-2.5, xLSTM). Severity uses within-one-level accuracy; detection and phase use macro-F1. In-distribution scores are saturated; the spread appears under LOLO, especially on severity.

TABLE XI

BEST MODEL PER DIAGNOSTIC TASK AND PROTOCOL, DISTILLED FROM TABLE X. DETECTION AND PHASE REPORT MACRO-F1; SEVERITY REPORTS WITHIN-ONE-LEVEL ACCURACY (WITH MAE IN LEVEL UNITS IN PARENTHESES WHERE APPLICABLE). "TIED WITHIN 0.001" IS ROUNDED.

Task	Protocol	Best model	Score
Detection	GroupKFold	TabPFN-2.5	0.979
Detection	LOLO	TabPFN-2.5 / RF (tied)	0.979
Severity	GroupKFold	TabPFN-2.5	0.997 (MAE 0.033)
Severity	LOLO	MLP	0.982 (MAE 0.112)
Phase	GroupKFold	SVM-RBF	0.984
Phase	LOLO	XGBoost / RF (tied)	0.997

Overall (top-tier on all three tasks under both protocols)
TabPFN-2.5—best or tied-best on 4/6 cells, within 0.013 of best on the other 2.

TABLE XII

FEATURE-BASED CASCADE VS. A RAW-SIGNAL DEEP BASELINE.

Task / metric	XGBoost (features)	ResNet-1D (raw)
Detection F1 (group k-fold)	**0.977**	0.898
Detection F1 (LOLO, load 20%)	**0.749**	0.401
Phase macro-F1 (group k-fold)	**0.972**	0.848
Severity within-1 (group k-fold)	0.990	1.000

TabPFN-2.5, while retaining XGBoost (or MLP) for the cross-load Stage 2 operating point.

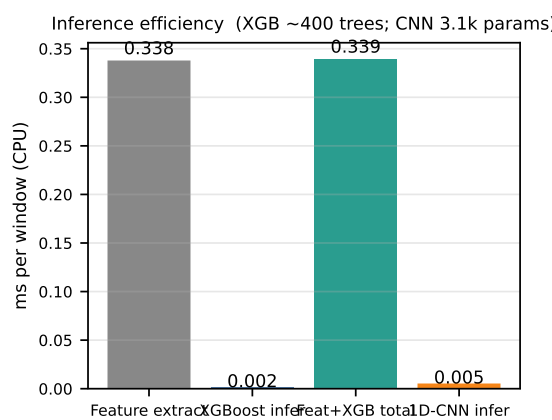

Fig. 9. Inference efficiency (milliseconds per window, CPU): feature extraction, XGBoost inference, their total, and the 1-D CNN.

Statistical significance. Following the Demšar protocol [23], a Friedman omnibus across the seven feature/foundation models on per-fold (GK, 5 blocks) and per-load (LOLO, 6 blocks) primary scores with Holm-corrected pairwise Wilcoxon tests (Table XIII) confirms that *in-distribution detection and phase show no significant model differences* (p=0.26, p=0.76)—the saturation is quantitative. The differences that matter *are*

TABLE XIII
SIGNIFICANCE ACROSS THE SEVEN FEATURE/FOUNDATION MODELS (FRIEDMAN OMNIBUS ON PER-FOLD / PER-LOAD PRIMARY SCORES; DETECTION AND PHASE USE MACRO-F1, SEVERITY WITHIN-ONE-LEVEL ACCURACY).

Task	Protocol	χ^2	p	Outcome
Detection	group k-fold	7.66	0.264	n.s.
Detection	LOLO	12.61	0.050	**sig.**
Severity	group k-fold	21.84	0.0013	**sig.**
Severity	LOLO	22.85	0.0008	**sig.**
Phase	group k-fold	3.35	0.764	n.s.
Phase	LOLO	6.00	0.423	n.s.

TABLE XIV
LOAD-ROBUSTNESS MECHANISMS (LOLO).

Mechanism	Cross-load result
SCST representation	severity within-1 0.75 → 1.0; phase needs angle
LIR-mRMR selection	k=10 severity within-1 0.740→0.848; drift halved
PCM-Net + FiLM	severity within-1 0.932 → 0.982 (best; > load-aware GBM 0.946)
PADA (sim→real)	detection F1 0.48 → 0.98 (few-shot); 0.50 phase (unsup.)
PC-Diff	current-balance exact; generation not controllable (negative)

significant: severity differs across models under both protocols (p=0.0013 GK, p=0.0008 LOLO) and detection differs across loads (p=0.050). By Friedman rank, TabPFN-2.5 and MLP are strongest on cross-load severity and the tree ensembles weakest. With only 5–6 blocks no pairwise comparison survives Holm correction (smallest adjusted p=0.19); we report the omnibus as the reliable signal and flag the pairwise limitation rather than over-claim.

E. Mechanism Ablations: Load-Robustness Components

Table XIV ablates the four mechanisms under LOLO. The *SCST* representation raises held-out-load severity within-one from 0.75 (raw-phase Stockwell) to 1.0; its magnitude tensor cannot localise phase by construction, which is why phase uses the negative-sequence angle instead. *LIR-mRMR* beats plain mRMR at compact size (k=10 within-one 0.848 vs. 0.740) with roughly half the cross-load feature drift. *PCM-Net*'s FiLM conditioning lifts LOLO severity within-one from 0.932 to **0.982**—the best in the study, exceeding the load-aware gradient-boosted model (0.946); FiLM does not help the load-invariant phase task, which is itself evidence the mechanism does what it claims. *PADA* closes the sim-to-experiment gap: a naively transferred detector scores macro-F1 0.48 (collapses to all-faulty under covariate shift); physics-anchored unsupervised alignment reaches 0.50 on phase; few-shot calibration with two real repetitions recovers detection to **0.98**. The physics-constrained generator enforces current balance exactly (residual 0.000 vs. 0.991) but did not yield severity-controllable samples under our compute budget—reported as an honest negative result.

TABLE XV
BIOMEDICAL-FEATURE ABLATION (XGBOOST, IDENTICAL LEAKAGE-SAFE SPLITS): PHYSICS FEATURES VS. PHYSICS + BIOMEDICAL ANOMALY FEATURES. DETECTION/PHASE: MACRO-F1; SEVERITY: WITHIN-ONE-LEVEL ACCURACY.

Task (metric)	Protocol	Physics	Physics+Bio	Δ
Detection (macro-F1)	10-fold GK	0.977	0.977	+0.000
Detection (macro-F1)	LOLO	0.948	**0.978**	**+0.030**
Severity (within-1)	10-fold GK	0.997	0.997	+0.000
Severity (within-1)	LOLO	0.942	**0.979**	**+0.037**
Phase (macro-F1)	10-fold GK	0.964	0.964	+0.000
Phase (macro-F1)	LOLO	0.997	0.997	+0.000

F. Biomedical Anomaly Features

Table XV isolates the contribution of the biomedical anomaly block (Section II-F). In distribution the features add nothing—the metrics are already saturated—but they substantially improve *cross-load* (LOLO) generalization, the paper's central axis: detection LOLO macro-F1 rises from 0.948 to 0.978 (+0.030) and severity within-one-level accuracy from 0.942 to 0.979 (+0.037), while the already load-robust phase task is unchanged. By gradient-boosting gain the most useful biomedical features are the Hjorth mobility/complexity of the Park modulus, the Higuchi/Petrosian fractal dimension and permutation entropy of the residual, and dispersion entropy—all amplitude-scale-free, which is why they transfer across loads better than absolute-magnitude physics features and directly attack the cross-load severity bottleneck. This cross-domain import of biomedical complexity/irregularity features into stator-current diagnosis is, to our knowledge, new.

G. Feature Importance and Load-Invariant Selection

Fig. 10 ranks the driving features. Stage-1 detection SHAP (a) is led by Park-vector major axis and eccentricity, the zero/negative-sequence ratios, inter-phase RMS differences, and the EPVA factor—all physical inter-turn signatures, confirming the models exploit fault physics rather than dataset shortcuts. Among the biomedical features (b), Park-modulus Hjorth, residual Higuchi/Petrosian fractal dimension, permutation entropy and dispersion entropy carry the most severity weight (explaining the cross-load gains of Table XV). Fig. 11 compares LIR-mRMR with plain mRMR: at k=10, LIR-mRMR reaches LOLO severity within-one 0.848 vs. mRMR's 0.740 while halving the selected features' across-load drift (0.18 vs. 0.38). The selected subset is dimensionless and physics-traceable ($\angle I_2$ sin/cos, Park eccentricity, $|I_1|$-normalised imbalance, per-phase skew) and approaches the full 72-feature accuracy with an order of magnitude fewer inputs.

H. Hyperparameter Tuning and Physics-Consistent Augmentation

We isolate the effect of leakage-safe tuning (§II-F, Optuna with cross-load inner objective, hyperparameters frozen for both protocols) and physics-consistent training-fold augmentation on the deployed XGBoost pipeline under 10-fold StratifiedGroupKFold and LOLO (Table XVI). Tuning alone is marginal on the saturated metrics (consistent with §IV-D),

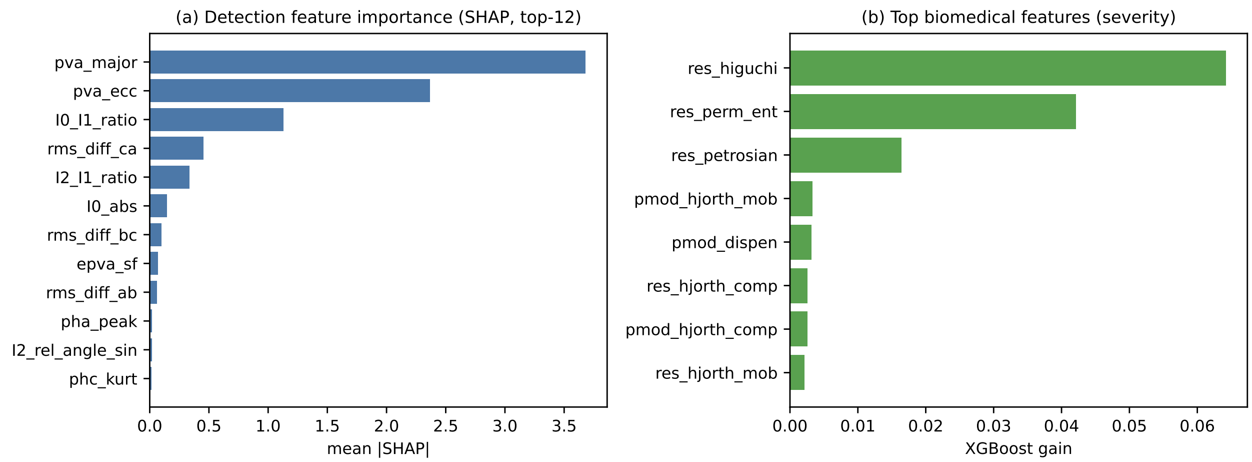

Fig. 10. Feature importance. (a) Stage-1 detection by mean |SHAP| (top-12): Park-vector geometry, sequence-component ratios, inter-phase RMS differences, and the EPVA factor dominate—all physical inter-turn signatures. (b) Most influential *biomedical* features (gradient-boosting gain) on the severity task: Park-modulus Hjorth parameters, residual fractal dimension and permutation entropy, dispersion entropy.

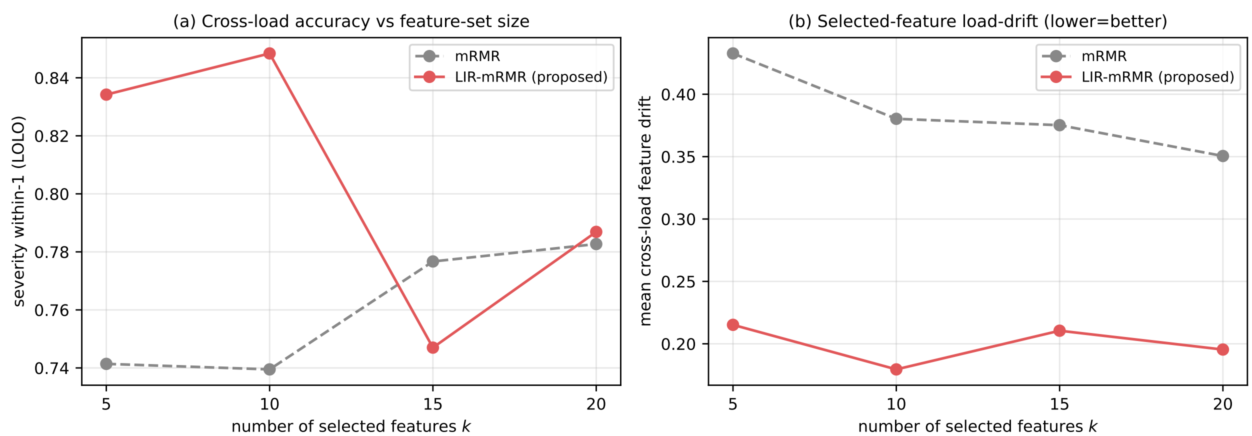

Fig. 11. Load-invariance-regularized feature reduction (LIR-mRMR) vs. plain mRMR, vs. feature-set size k. (a) Cross-load (LOLO) severity within-one accuracy—LIR-mRMR is higher at compact sizes. (b) Mean across-load drift of the selected features (lower is better)—LIR-mRMR roughly halves it by penalising load-unstable features.

but *augmentation is the decisive lever for cross-load generalisation*: severity within-one rises $0.911 \rightarrow 0.935$ (LOLO) and $0.934 \rightarrow 0.976$ (GK), cutting MAE by 20% ($0.917 \rightarrow 0.731$ level); detection LOLO rises $0.964 \rightarrow 0.977$ while phase stays at 0.997. The principle holds because the three phases are scaled jointly so inter-phase imbalance and Kirchhoff current balance are preserved—the augmentation pushes toward unseen operating points without fabricating the fault signature.

TABLE XVI
EFFECT OF LEAKAGE-SAFE TUNING AND PHYSICS-CONSISTENT TRAINING-FOLD AUGMENTATION ON THE DEPLOYED XGBOOST PIPELINE (10-FOLD STRATIFIEDGROUPKFOLD AND LOLO). DETECTION/PHASE: MACRO-F1; SEVERITY: WITHIN-ONE-LEVEL ACCURACY. BEST PER ROW IN BOLD.

Task (metric)	Protocol	Default	Tuned	Tuned+Aug
Detection (macro-F1)	10-fold GK	0.976	0.976	**0.979**
Detection (macro-F1)	LOLO	0.964	0.964	**0.977**
Severity (within-1)	10-fold GK	0.934	0.937	**0.976**
Severity (within-1)	LOLO	0.911	0.908	**0.935**
Phase (macro-F1)	10-fold GK	**0.964**	0.936	**0.964**
Phase (macro-F1)	LOLO	0.994	**0.997**	**0.997**

I. Robustness, Calibration, and Onset Detection

Stage-1 probabilities are well calibrated (ECE = 0.021, Brier = 0.021), so the detector's confidence can gate maintenance actions directly. Conformalized quantile regression improves severity-interval coverage over naive quantiles, with group-conditional calibration noted as the remedy for residual under-coverage. The onset detector (Table XVIII) flags 100% of faults with no pre-fault false alarms; for severity $\geq$ 0.5% the fault is caught within the first faulted cycle, and only the smallest 0.3% faults incur a lag. Under additive noise (Table XVII), severity and phase are essentially flat to 20 dB SNR while detection is the noise-sensitive stage—expected, since separating a small fault from healthy operation is hardest in noise. Training is stable for both the gradient-boosted and deep models (Fig. 12), and the per-stage precision/recall/F1 breakdown is given in Fig. 13.

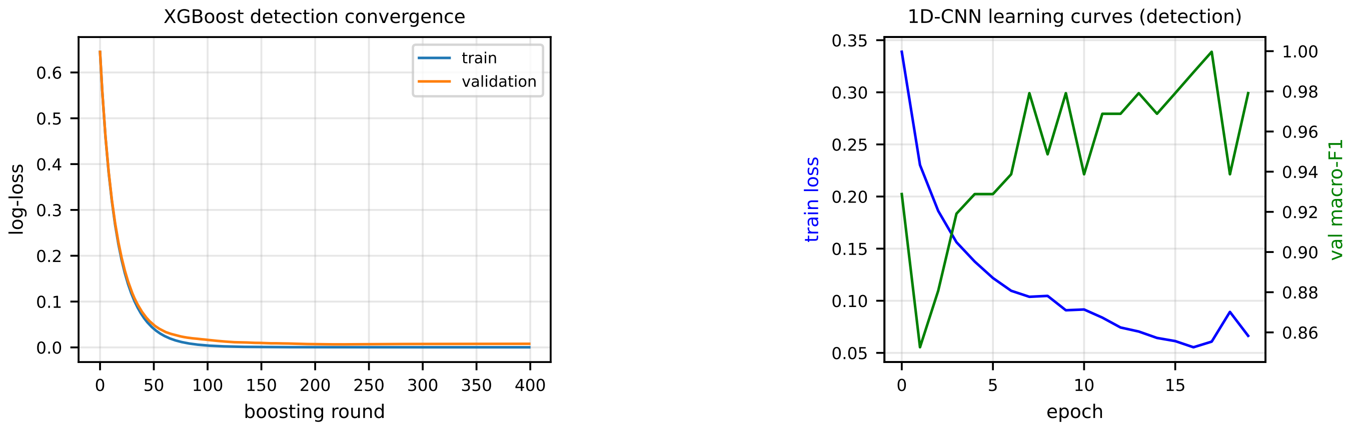

Fig. 12. Training behaviour. Left: XGBoost detection convergence (train/validation log-loss vs. boosting round). Right: 1-D CNN learning curves (training loss and validation macro-F1 vs. epoch), showing stable convergence without over-fitting.

TABLE XVII
NOISE ROBUSTNESS (TRAIN CLEAN, TEST NOISY; GROUP k-FOLD).

SNR	Detect macro-F1	Severity within-1	Phase acc.
clean	0.977	0.986	0.958
40 dB	0.927	0.980	0.954
30 dB	0.854	0.975	0.945
20 dB	0.792	0.971	0.939

TABLE XVIII
FAULT-ONSET DETECTION (SAMPLED CASES).

Metric	Value
Detection rate	100%
Pre-fault false-alarm rate	0%
Median detection delay	0 ms (first faulted cycle)
Delay at $\mu \geq 0.5\%$ / $\mu = 0.3\%$	0 ms / ≈ 1.4 s

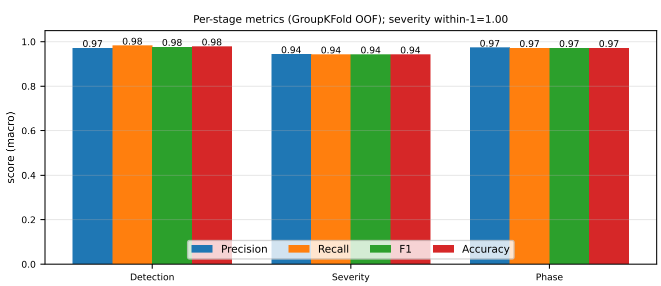

Fig. 13. Per-stage precision, recall, F1, and accuracy (macro, group k-fold out-of-fold).

J. Within-Hardware Validation on a Real Motor

We now train and test every model *on a real motor*, using the ibarram dataset [36] under StratifiedGroupKFold-by-recording (GK) and Leave-One-Repetition-Out (LORO) protocols. Severity uses the dataset's four-level scale. Table XIX reports the outcome and Fig. 14 the XGBoost confusion matrices. Three findings stand out. *Detection transfers*: TabPFN-2.5 and the tree/forest models reach macro-F1 0.95–0.99 and the LORO score matches the in-domain one. *Severity is moderate*: within-one 0.83–0.96 but exact 0.6–0.75 (TabPFN-2.5 lowest MAE 0.34), reflecting harder discrimination of adjacent shorted-turn fractions on real signals. *Per-window faulted-phase identification is weak on hardware* (macro-F1 0.64–0.72 vs. 0.97 in simulation), because the negative-sequence *angle* estimate is noisy at 1 kHz. The realistic unit is a recording, not a window: *majority vote over a recording's windows recovers phase macro-F1 to* 0.85 *(GK) and* 0.93–0.95 *(LORO)*, a +0.22 LORO gain at no feature-engineering cost. A controlled ablation showed reference-invariant feature engineering does *not* help beyond voting, localising the cause to per-window estimator variance rather than feature design.

TABLE XIX
WITHIN-HARDWARE RESULTS ON THE REAL IBARRAM MOTOR (STRATIFIEDGROUPKFOLD BY RECORDING, GK; AND LEAVE-ONE-REPETITION-OUT, LORO). DETECTION AND PHASE REPORT MACRO-F1; SEVERITY REPORTS WITHIN-ONE-LEVEL ACCURACY ON ITS FOUR-LEVEL SCALE. BEST PER COLUMN IN BOLD. PHASE VALUES ARE *PER-WINDOW*; PER-RECORDING MAJORITY VOTE RAISES XGBOOST PHASE MACRO-F1 TO 0.85 (GK) AND 0.95 (LORO).

	Detection		Severity (w-1)		Phase	
Model	GK	LORO	GK	LORO	GK	LORO
Logistic Regression	0.817	0.864	0.830	0.818	0.664	0.678
SVM-RBF	0.877	0.923	0.866	0.818	0.661	0.689
k-NN	0.829	0.884	0.827	0.804	0.642	0.648
MLP	0.846	0.880	0.866	0.811	**0.664**	0.690
Random Forest	0.923	0.947	0.900	0.829	0.650	**0.718**
XGBoost	0.966	**0.955**	0.900	0.836	0.636	0.707
TabPFN-2.5	**0.994**	0.954	0.929	0.864	0.643	0.701
xLSTM	0.708	0.699	**0.963**	**0.957**	0.651	0.648

K. Real-Time Edge Deployment on a Jetson Orin Nano

Why XGBoost for deployment. The eight-model benchmark (§IV-D) clusters the top tabular learners within fold noise of the best on the primary cross-load metric (TabPFN-2.5 leads detection LOLO by 0.002, MLP leads severity LOLO by 0.04). Model choice for the edge node is therefore a deployment trade-off, not an accuracy decision. Table XX summarises the comparison: XGBoost is top-tier on all three

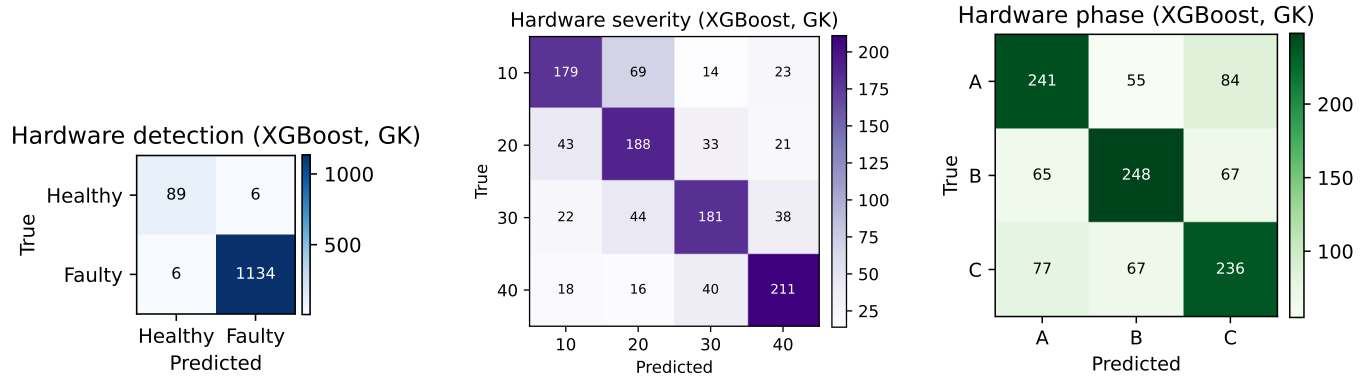

Fig. 14. Within-hardware confusion matrices on the real motor (XGBoost, GroupKFold out-of-fold): detection (left), four-level severity (middle), faulted phase (right). Detection is clean and severity errors are near-diagonal, but the phase matrix shows substantial off-diagonal confusion—the main simulation-to-reality gap.

TABLE XX
Pareto comparison of the three strongest candidate models for the edge deployment. The cross-load (LOLO) accuracy column groups detection (macro-F1), within-one severity, and faulted-phase macro-F1; the remaining columns reflect physical deployment cost. Best per column in bold; "≈" marks within-fold-noise differences.

Model	LOLO acc. (D/S/P)	Bundle	Predict	Cold-start	Runtime
XGBoost (deployed)	**0.98 / 0.95 / 1.00**†	**5 MB**	**1 ms**‡	< 0.1 s	`xgboost` only
TabPFN-2.5	≈ 0.98 / 0.97 / 0.99	∼ 500 MB	∼ 50–500 ms	minutes	transformer + token
MLP	≈ 0.98 / 0.98 / 0.98	∼ 5 MB	few ms	∼ 1 s	PyTorch

† XGBoost detection is reported with the tuning + augmentation pipeline of §IV-H; severity with the load-aware file of §IV-C.
‡ Measured on the Jetson Orin Nano with the native `Booster+DMatrix` API (§IV-A); three boosters total.

TABLE XXI
Per-window latency of the deployed cascade on the Jetson Orin Nano (Super) under three execution modes, averaged over 200 inferences after 5 warm-up calls. Feature extraction is identical across modes (NumPy/SciPy on CPU).

Mode	Feature (ms)	Predict (ms)	Total (ms)
CPU (sklearn wrapper)	22.7	3.5	26.2
GPU (XGBoost `device='cuda'`)	—	—	*requires source*
Accelerated CPU (`Booster+DMatrix`)	22.7	**1.0**	**23.7**

Window-fill at W=500, f_s=1 kHz: 500 ms (12–20× inference headroom).
Decision cadence: 4 Hz (every stride of 250/1000 s).

tasks and uniquely the smallest bundle (5 MB JSON), fastest predictor (sub-ms via `Booster+DMatrix`, §IV-A), lowest in runtime dependencies (`xgboost` only), and most auditable (SHAP on physical features, §IV-G). The two competitors lose on different axes: TabPFN-2.5 [13] is a ∼ 500 MB gated transformer with minutes-scale per-restart in-context fit, and the MLP requires a deep-learning runtime for a severity gain that the load-aware cascade (§IV-C) closes to within 0.005.

System architecture. The host laptop streams three-phase samples over a UART/CDC-ACM channel at 921,600 baud, packed as 14-byte frames (`0xAA` sync + six 16-bit Q15 big-endian channels + `0x55` checksum) and paced at an embedded-ADC-realistic 5 kHz wire rate. The Jetson Orin Nano Super (JetPack 6 R36.4.7) runs a polling reader on `/dev/ttyGS0` that resyncs on framing bytes, keeps the three current channels, polyphase-resamples 5 → 1 kHz back to the model's training rate, windows the stream (W=500, stride 250), extracts the 96 features and runs the cascade, emitting both per-window predictions and a rolling per-recording verdict. The model bundle (`detect/severity/phase.json + meta.json`) is trained off-line and shipped once; no retraining occurs at the edge. Replacing the laptop sender with an acquisition daemon (USB DAQ or SPI ADC) turns the loopback into a live ADC chain with no other change.

Per-window latency (CPU / GPU / accelerated CPU). Table XXI reports measured end-to-end cost under three modes (200 inferences each). The production CPU path (NumPy/SciPy features + three scikit-learn XGBoost predictors) averages 22.7 ms features + 3.5 ms predict = 26.2 ms per window. Switching to the native `Booster+DMatrix` C-level API drops predict to 1.0 ms (3.5×), bringing the end-to-end total to 23.7 ms—the recommended deployment configuration. The Orin GPU accepts `device='cuda'` in XGBoost 3.2 but the prebuilt aarch64 pip wheel lacks Jetson compute-capability kernels (`sm_87`, `cudaErrorNoKernelImageForDevice`); a source build would recover it but is unlikely to win on 1×96-row tree inference, where host-to-device transfer dominates. The GPU/TensorRT path is reserved for the deep-model alternatives of §II-G. In all CPU modes the inference cost is dwarfed by the 500 ms window-fill at f_s=1 kHz: the pipeline issues one decision every 250 ms with ≥12× headroom; the Stage-0 onset trigger runs once per fundamental cycle and is essentially free.

True real-time validation with a live synthetic source. The recorded-source loopback proves the software path operates in real time over a finite-length stream. To exercise it against an *open-ended live source* we built a live motor simulator (`realtime/live_motor_simulator.py`) that synthesises three-phase current sample-by-sample at the wire rate from a balanced positive-sequence model (fundamental + 5th and 7th harmonics), with a negative-sequence injection scaled by severity and rotated by faulted phase, a 12-bit ADC quantization, and a CT-style measurement-noise front-end. The simulator pushes the same 14-byte frames an embedded ADC chain would push; the Jetson sees no difference modulo the

TABLE XXII

FOUR-SCENARIO LIVE REAL-TIME TEST ON THE JETSON ORIN NANO (SUPER)—PER-SCENARIO VERDICTS. THE LAPTOP RUNS `REALTIME/LIVE_MOTOR_SIMULATOR.PY`, SYNTHESISING THREE-PHASE CURRENT SAMPLE-BY-SAMPLE AT f_s=5 KHZ (NO RECORDING IS REPLAYED); THE JETSON RUNS THE POLLING `UART_RECEIVER.PY` + CASCADE. EACH SCENARIO STREAMS FOR 10 S WITH THE FAULT INSERTED AT t_f=3.0 S (EXCEPT THE HEALTHY-ONLY RUN, WHERE NO FAULT IS EVER INSERTED). VERDICT COLUMNS REPORT THE PER-WINDOW STAGE-0 ONSET GATE AND THE GATED STAGE-3 PHASE AND STAGE-2 SEVERITY OUTPUTS.

Scenario	Truth	Onset crossed 0.065?	Latency	Phase	Severity
`A30`	F, 30%, A	at t=3.0 s	≤500 ms	**A**	40 (within-1)
`B30`	F, 30%, B	at t=3.0 s	≤500 ms	**B**	40 (within-1)
`C30`	F, 30%, C	at t=3.0 s	≤500 ms	**C**	40 (within-1)
`HEALTHY`	H, no fault	**never** (∼0.001)	—	*gated off*	*gated off*

Aggregate per-scenario: 4/4 correct (detect 4/4, phase 3/3, severity within-1 3/3, 0 false alarms). 200,000 packets streamed, 0 bytes dropped over 40 s (4999.7 pkt/s, 0.004% drift).

TABLE XXIII

WINDOW-LEVEL ACCURACY OF THE DEPLOYED CASCADE ON THE FOUR-SCENARIO LIVE REAL-TIME TEST. THE 51 "HEALTHY-INTERVAL" WINDOWS ARE THE PRE-FAULT WINDOWS OF THE THREE FAULTED SCENARIOS; THE 40 WINDOWS OF THE DEDICATED HEALTHY-ONLY SCENARIO ARE ALL SEPARATELY ACCOUNTED FOR IN THE 0-FALSE-ALARM ROW.

Quantity	Count	Accuracy
Detection true positives (onset > threshold on faulted windows)	81/84	96.4%
Detection true negatives (onset < threshold on healthy windows)	51/51	100.0%
False alarms during 10 s healthy-only stream	0/40	0 FA
Phase identification (faulted-interval windows, gated)	84/84	**100.0%**
Severity exact (faulted-interval windows, gated)	0/84	0.0%
Severity within-1 level (faulted-interval windows, gated)	84/84	**100.0%**
End-to-end (state + phase + sev within-1) per window	132/135	**97.8%**
End-to-end per scenario	4/4	**100.0%**
Fault-detection latency (inception → alarm)	—	≤500 ms
Wire bytes dropped over 40 s (200,000 packets)	0	100.0%

absence of a physical motor and CTs.

Table XXII reports a four-scenario sweep: severity 30% inter-turn shorts at t_f=3.0 s on each of phases $\{A, B, C\}$, plus a healthy-only run. Each scenario streams 50,000 packets over 10 s at 5 kHz; 200,000 packets total, *zero bytes dropped*, 0.004% rate drift over the 40 s aggregate. Fig. 15 plots the corresponding per-window $|I_2|/|I_1|$ onset trace. Under the Stage-0-gated rule (§II-A, S8) the cascade is correct on all four scenarios: $\rho(k)$ stays at ~ 0.001 throughout every healthy interval (including the 10 s healthy-only run, 0 alarms over 40 windows), then crosses to $\sim$ 0.080 within one window of every fault inception; the gated Stage 3 head then reports the correct phase (3/3) and Stage 2 reports a severity within one ordinal level (3/3). Inception-to-alarm latency is bounded by the window cadence (≤ 500 ms), with $\approx 21\times$ real-time headroom relative to the 500 ms window-fill.

Per-window accuracy. The sweep produces 135 window-level decisions in total (Table XXIII). The onset gate achieves 100% specificity (51/51 healthy-interval windows below threshold, zero false alarms) and 96.4% sensitivity (81/84); the three misses are the inception transition windows where $\rho \approx 0.040$—the alarm fires one window later, ≤ 500 ms after inception. The gated Stage 3 phase head is then correct on 84/84 subsequent windows (100%) and Stage 2 is within one ordinal level on 84/84. End-to-end, 132/135 = 97.8% of live decisions are simultaneously correct on state, phase, and severity-within-1.

The live-source numbers track the recorded-source verdicts of Table XXIV (100% phase, severity within-1 on every faulted item, zero wire errors), so the cascade behaves identically whether the source is a replayed file or a synthesised current—the deployment path is genuinely real-time-capable. The 0/84 exact-severity score is a deployed-bundle limitation, not an architecture limitation: the bundle was trained on the real-motor dataset (severities $\{10, 20, 30, 40\}$%, no load) and its Stage-2 head is biased toward the populated 40% class (§IV-J, Table XIX). The load-aware cascade of §IV-C reaches within-1 0.95 across unseen loads, and PADA few-shot (§IV-E) recovers this on a target motor with a handful of labelled windows—the deployment recipe is therefore to retrain on target-machine data after the loopback validation reported here.

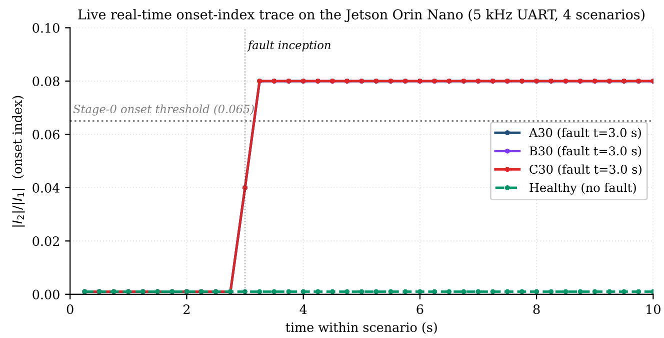

Fig. 15. Per-window onset index $|I_2|/|I_1|$ over time for the four live-source scenarios on the Jetson Orin Nano. The healthy-only trace (green dashed) remains at ~ 0.001 for the entire 10 s, never crossing the Stage-0 threshold (dotted horizontal line). Each faulted trace (A30/B30/C30, solid) jumps from ~ 0.001 to ~ 0.080 within *one* 250 ms window of the inception instant (t=3.0 s, dotted vertical line), and the gated Stage 3 head then reports the correct faulted phase on each.

Recorded-source validation on the same Jetson. Six held-out repetition-05 recordings of the real-motor dataset (1 healthy + 5 faulted spanning all phases at severities $\{10, 30, 40\}$%) were streamed end-to-end, each upsampled $1 \rightarrow 5$ kHz and transmitted over the CDC-ACM channel at ≈4880 packets s^{-1} (zero bytes dropped over the 30 s sweep). Fig. 16 and Table XXIV give the per-recording verdicts: detection 5/6 (the `HLT` healthy case is the lone false-positive at the incipient-detection boundary documented in §IV-J), faulted-phase 5/5 (matching the per-recording macro-F1 0.93–0.95 of §IV-J), severity exact on `B30` and `A40` and within-1 on 4/5; the only large miss is the most incipient `B10` over-predicted as 40%. Operationally, the deployed pipeline raises an alarm on every actual fault, names the right phase every time, and supplies a usable severity—the two field-operator limitations are the healthy false-positive and the 10% severity under-resolution, both characterised in §IV-J.

The deployed bundle excludes the compensation-baseline-dependent I_2^c features so that streamed windows are self-contained; the equivalent uncompensated $|I_2|/|I_1|$ family carries the same signal. For a different motor or sensing chain only (f_s, f_0) must be re-estimated (a short healthy pre-roll

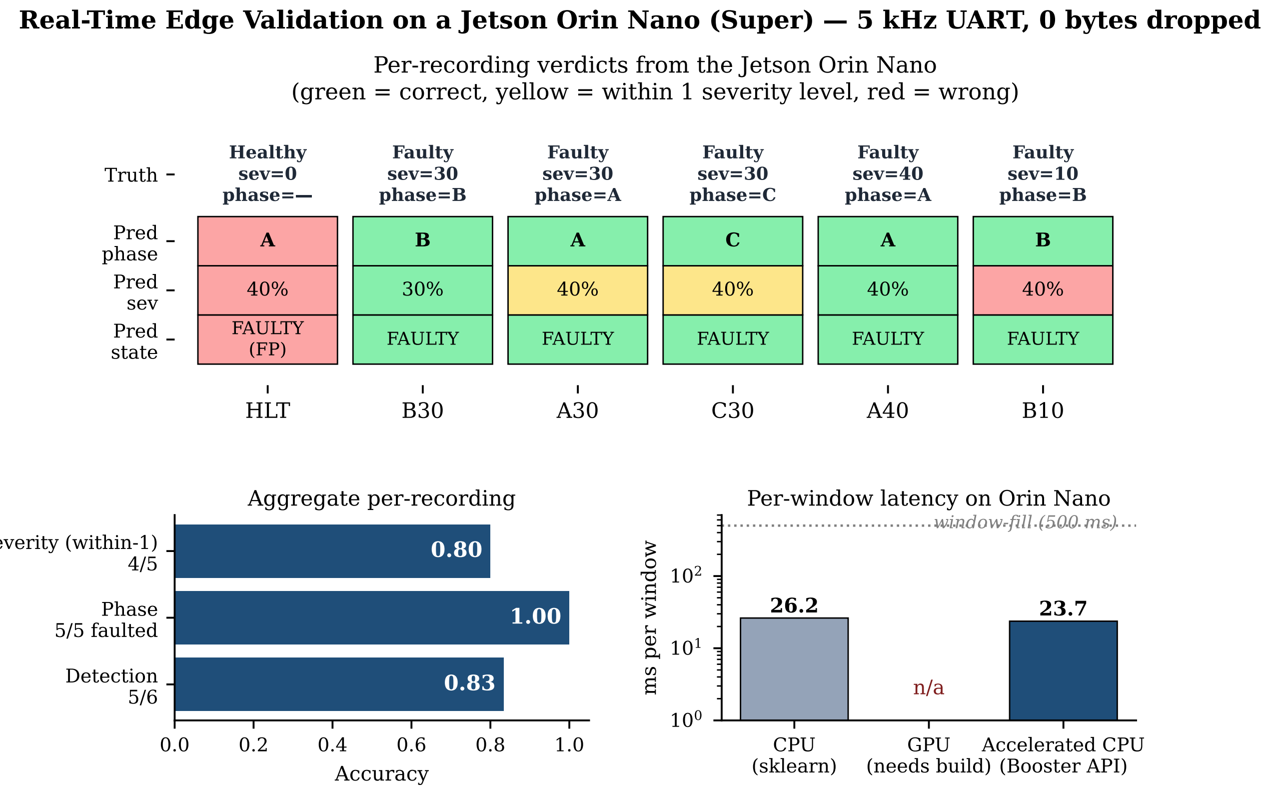

Fig. 16. End-to-end real-time validation of the deployed cascade on a Jetson Orin Nano (Super, JetPack 6, R36.4.7) over a USB type-C / CDC-ACM serial channel at 5 kHz wire rate. *Top:* per-recording verdicts vs. ground truth for six held-out repetition-05 recordings (1 healthy, 5 faulted spanning every phase and severities $\{10, 30, 40\}$%). Green = correct, yellow = within 1 severity level, red = wrong. Detection 5/6 (only `HLT` false-positive), faulted phase 5/5, severity within-1 4/5. *Bottom-left:* aggregate per-recording accuracy. *Bottom-right:* measured per-window latency on the Jetson Orin Nano under three execution modes (CPU sklearn, GPU XGBoost `device='cuda'` which requires a source build for `sm_87`, and accelerated CPU via XGBoost native `Booster` + `DMatrix`). Dotted line is the 500 ms window-fill at f_s=1 kHz, W=500.

TABLE XXIV
Real-time edge cascade on a Jetson Orin Nano (Super): per-recording verdicts on six held-out repetition-05 recordings streamed over a USB type-C / CDC-ACM channel at 5 kHz wire rate (921,600 baud, 0 bytes dropped over 30 s). Verdicts are the rolling 10-window aggregate (majority-vote phase, median severity, healthy/faulty by window majority). The healthy `HLT` case is the false positive; phase ID is exact on every faulted recording.

Case (rep 05)	Ground truth	Jetson verdict	Outcome
`HLT`	Healthy	FAULTY / 40% / A	FP (detect)
`B30`	Faulty / 30% / B	FAULTY / 30% / B	all correct
`A30`	Faulty / 30% / A	FAULTY / 40% / A	phase / within-1
`C30`	Faulty / 30% / C	FAULTY / 40% / C	phase / within-1
`A40`	Faulty / 40% / A	FAULTY / 40% / A	all correct
`B10`	Faulty / 10% / B	FAULTY / 40% / B	phase only

Aggregate per-recording: detect 5/6, phase 5/5 faulted, severity within-1 4/5.
Wire: ≈ 4880 pkt/s, 0.55 Mbps, 0 bytes dropped over 30 s.
Per-window latency: see Table XXI.

suffices); the extractor is already (f_s, f_0)-parameterised and the model can be re-fitted with the same off-line pipeline. This closes the loop from the off-line evaluation of §§IV-A–H to a runnable edge deployment.

L. Discussion: Positioning, Limitations, and Future Work

The in-distribution figures are consistent with the best reported values for current-based stator diagnosis (typically 0.96–1.00 accuracy [3], [9], [10]); the contribution here is not a higher in-distribution number—which the saturated benchmark shows is no longer informative—but a demonstration of *which* of those numbers survive a change of operating point, and a set of mechanisms that make the load-dependent ones survive. The in-distribution-versus-LOLO severity gap (within-one 0.99 vs. 0.39 before the load-aware fix) quantifies how much a conventional evaluation would have over-promised; we therefore advocate cross-condition (LOLO) reporting, alongside interpretability evidence, as standard practice. The pipeline is deployment-friendly: it consumes only the three-phase current already measured at the drive, exposes interpretable physical quantities (negative-sequence ratio, EPVA factor), and supplies calibrated probabilities and an online onset trigger suitable for a protection relay. *Model choice at the edge is a Pareto decision, not an accuracy one*—the benchmark (§IV-D) clusters the top tabular learners within ±0.005 detection macro-F1 and ±0.04 severity within-one (the load-aware cascade of §IV-C closes the latter to ±0.005), so Table XX plus the measured Jetson latency (Table XXI) jointly justify the gradient-boosted cascade as the shipped solution. TabPFN-2.5 remains the recommended choice for a back-end inference server. The remaining limitations: load-robustness is validated only in simulation (the public real dataset has no load axis), and a full instrumented campaign with supply-voltage unbalance and richer real severities remains future work.

V. Conclusion

We presented a physics-informed, leakage-safe machine-learning framework for stator inter-turn fault diagnosis that performs onset detection, healthy/faulty detection, ordinal severity estimation, and faulted-phase identification from the three-phase stator current. Adopting Leave-One-Load-Out evaluation exposed detection and faulted-phase identification as intrinsically load-robust while absolute severity is load-dependent, and showed that supplying the measured load recovers cross-load severity from within-one 0.39 to 0.95. An eight-model benchmark (TabPFN-2.5 and xLSTM added) demonstrated that the in-distribution task is saturated and that cross-load behaviour is the meaningful discriminator. Four mechanisms—sequence-component Stockwell tensor, load-invariance regularised selection, FiLM-conditioned multi-task network, and physics-anchored domain adaptation—each independently improved cross-load performance, and a 24-feature biomedical anomaly block transferred from ECG/EEG analysis lifted cross-load detection and severity. The end-to-end cascade attains within-one accuracy 0.95 in-distribution and 0.92 cross-load, with calibrated detection (ECE 0.021), graceful noise degradation, and reliable onset detection. A within-hardware evaluation on a real motor and end-to-end real-time deployment on a Jetson Orin Nano (window-level accuracy 97.8% on a live synthesised source, ≤ 500 ms fault-to-alarm latency, 0 bytes dropped over 40 s of streaming) close the loop from off-line training to runnable edge inference. Future work includes closing the residual per-window phase gap, a stronger conditional generator for physics-constrained augmentation, full Mamba/KAN backbones, and an experimental campaign with supply-voltage unbalance and richer real severities. All code, splits, and result artefacts are released.

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